The Crop Journal 13 (2025) 5–22

Contents lists available at ScienceDirect

The Crop Journal

journal homepage: www.keaipublishing.com/en/journals/the-crop-journal/

Functions of transcription factor superfamilies in rice immunity

Sang Ryeol Park [a,1], Yujeong Jeong [a,b,1], Seungmin Son [a,*]

[a] National Institute of Agricultural Sciences, Rural Development Administration, Jeonju 54874, Republic of Korea
[b] Department of Crop Science and Biotechnology, Jeonbuk National University, Jeonju 54896, Republic of Korea

ARTICLE INFO

Article history:
Received 10 July 2024
Revised 10 October 2024
Accepted 15 October 2024
Available online 17 November 2024

Keywords:
Biotic stress
Crop resilience
Immune response
Rice
Transcription factor

ABSTRACT

The emergence of novel phytopathogens and the accelerated spread of plant diseases to new regions, driven by global climate change, constitute significant threats to agricultural resources. Rice, a major tropical staple crucial for global food security, possesses six transcription factor superfamilies—AP2/ERF, bHLH, bZIP, MYB, NAC, and WRKY—that function in innate immunity against pathogens. We review their biological functions and regulatory mechanisms in rice immunity.

© 2024 Crop Science Society of China and Institute of Crop Science, CAAS. Production and hosting by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Contents

1. Introduction . . . 5
2. AP2/ERF transcription factor in rice immunity . . . 7
3. bHLH transcription factor in rice immunity . . . 7
4. bZIP transcription factor in rice immunity . . . 9
5. MYB transcription factor in rice immunity . . . 12
6. NAC transcription factor in rice immunity . . . 12
7. WRKY transcription factor in rice immunity . . . 14
8. Concluding remarks and future perspectives . . . 17
CRediT authorship contribution statement . . . 18
Declaration of competing Interest . . . 18
Acknowledgments . . . 18
Appendix A. Supplementary data . . . 18
References . . . 18

1. Introduction

Climate change exacerbates the adverse impacts of biotic stress in the agricultural sector by increasing pathogen susceptibility and weakening plant defense mechanisms, thereby threatening food crops that are essential energy sources for the global population [1,2]. The intricate innate immunity mechanisms of plants, typified by pathogen-associated molecular pattern (PAMP)-triggered immunity (PTI) and effector-triggered immunity (ETI), are orchestrated by a myriad of signaling components and transcriptional networks. PAMP recognition by pattern recognition receptors activates PTI signaling pathway, such as mitogen-activated protein kinase (MAPK/MPK) cascade, resulting in basal defense responses [3]. However, while PTI can be subverted by pathogen effectors, resistance proteins (e.g., nucleotide-binding and leucine-rich repeat [NLR]) perceive pathogen effectors and trigger the robust immune defense ETI [4]. Transcription factors (TFs), including APETALA2/ethylene responsive factor (AP2/ERF), basic helix-loop-helix (bHLH), basic leucine zipper (bZIP), myeloblastosis (MYB), NAM,

* Corresponding author.
E-mail address: linewind@korea.kr (S. Son).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.cj.2024.10.006
2214-5141/© 2024 Crop Science Society of China and Institute of Crop Science, CAAS. Production and hosting by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

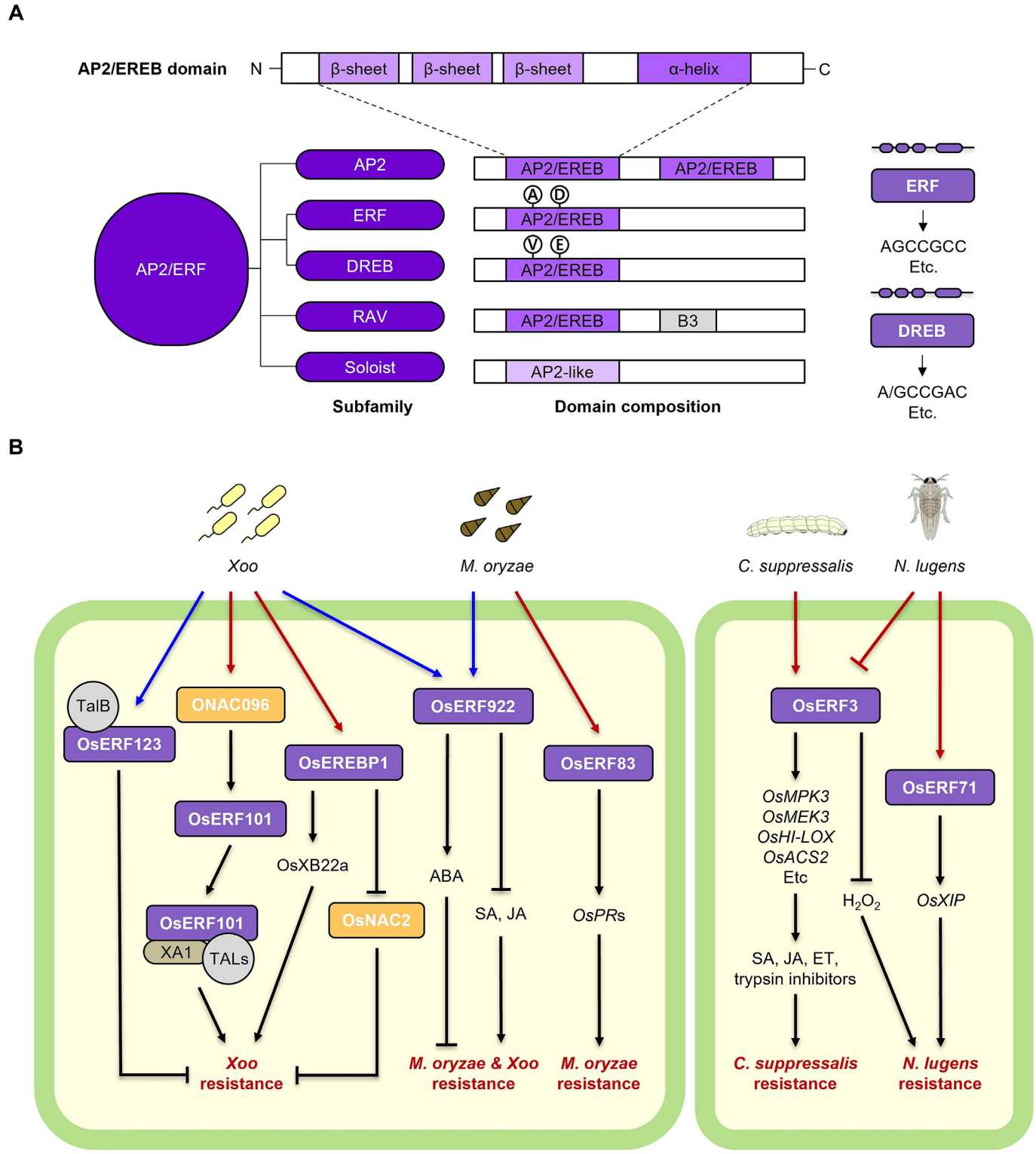


Fig. 1. Roles of AP2/ERF transcription factors in rice immunity. (A) Structural features of AP2/ERF subfamilies. The AP2/ERF superfamily is categorized into four subfamilies: AP2, ERF, RAV, and Soloist, based on domain composition. The ERF subfamily is further subdivided into ERF and DREB groups, distinguished by specific amino acids at residues 14 and 19. The ERF group binds to the ERE motif (core sequence AGCCGCC [GCC-box]) to mediate resistance to biotic stresses, whereas the DREB group binds to the DRE/CRT motif (core sequence A/GCCGAC) to confer tolerance to abiotic stresses. (B) Schematic illustration of the regulatory roles of AP2/ERF transcription factors in rice immunity against various pathogens. OsEREBP1, OsERF71, OsERF83, and OsERF101 positively regulate resistance to biotic stress. Conversely, OsERF922 and OsERF123 negatively regulate it. OsERF3 functions as both a positive and negative regulator of innate immunity against *C. suppressalis* and *N. lugens*, respectively. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility.

ATAF, and CUC (NAC), and WRKY DNA-binding protein (WRKY), mediate these processes.

Rice (genus *Oryza*), a primary dietary staple crop, is vulnerable to a broad spectrum of pathogens, including fungi (yield losses up to 100%), bacteria (yield losses up to 85%), viruses (yield losses up to 100%), and insect pests (yield losses of 20%–40%) [5,6], all of which cause annual yield reductions. Therefore, extensive knowledge of rice immunity against diverse pathogens is crucial for effectively preparing to mitigate this damage. A comprehensive genome-wide study has identified 3078 putative TFs in the rice genome, which have been categorized into 59 distinct families [7]. Of these, more than a third belong to six TF superfamilies: AP2/ERF, bHLH, bZIP, MYB, NAC, and WRKY. These TF superfamilies are responsive to various destructive pathogens, such as *Xanthomonas oryzae* pv. *oryzae* (*Xoo*) and *Magnaporthe oryzae*, and appear integral to rice immune responses [8–10]. Thus, to address the impending agricultural crisis, concerted attention to the six TF superfamilies is warranted. Here we review the biological functions and regulation mechanisms of these TF superfamilies in rice-pathogen interactions.

2. AP2/ERF transcription factor in rice immunity

The AP2/ERF superfamily, characterized by the presence of the conserved 60–70 amino acid AP2/EREB domain or AP2-like domain, comprises four subfamilies. These subfamilies are categorized based on the number of AP2/EREB domains and additional domains they contain (Fig. 1A): the AP2 subfamily typically contains two AP2/EREB domains; the ERF subfamily, further divided into ERF and DREB groups based on specific amino acids at residues 14 and 19, possesses one AP2/EREB domain; RAV subfamily harbors one AP2/EREB domain with a B3 domain; and the Soloist subfamily contains one AP2-like domain. The distinct DNA-binding specificities of AP2/ERF TFs are governed by specific amino acid residues within the AP2/EREB domain. ERF TFs, characterized by alanine at residue 14 and aspartic acid at residue 19, bind to ERE motif (core sequence AGCCGCC [GCC-box]), inducing resistance to biotic stresses [11,12], while DREB TFs, characterized by valine at residue 14 and glutamic acid at residue 19, bind to DRE/CRT motif (core sequence A/GCCGAC), thereby conferring tolerance to abiotic stresses [11,12].

The AP2/ERF superfamily regulates rice immunity against pathogens. The roles and mechanisms of rice AP2/ERF TFs in mediating responses to biotic stress are described in Fig. 1B and Table S1. The brown planthopper (BPH) *Nilaparvata lugens* and the striped stem borer (SSB) *Chilo suppressalis*, as piercing-sucking and chewing herbivores respectively, inflict substantial annual yield losses in rice. Defense responses against these insect pests involve the activation of MAPK and various phytohormone signaling pathways, including salicylic acid (SA), jasmonic acid (JA), and ethylene (ET). SA and JA increase resistance to piercing-sucking and chewing herbivores, respectively, potentially leading to an antagonistic effect [13,14]. OsERF3 increases resistance to SSB but reduces resistance to BPH [15]. The induction of OsERF3 by *C. suppressalis* upregulates various defense genes (*OsMPK3* and *MAPK KINASE 3* [*OsMKK3*/*OsMEK3*], *OsHI-LOX*/*OsLOX2;1*, and *OsACS2*), phytohormone biosynthesis (SA, JA, and ET), and the activity of trypsin protease inhibitors, resulting in increased immunity against SSB [15]. In contrast, OsERF3, which is repressed by *N. lugens* infection, suppresses H_2O_2 synthesis, leading to reduced resistance to this insect [15]. Thus, OsERF3 is a switch controlling rice immunity against both chewing and piercing-sucking insects. OsERF71 binds and activates the promoter of *OsXIP*, essential for resistance to BPH [16].

In vitro, BWMK1/OsMPK12 phosphorylates OsEREBP1/OsERF70, increasing its binding affinity to the GCC-box motif present in the promoter regions of several *PR* genes [17]. Since overexpression of *BWMK1* in tobacco and rice resulted in elevated expression of various *PR* genes and conferred broad-spectrum disease resistance (BSDR) [17,18], it was suggested that OsEREBP1 is involved in MAPK signaling pathway associated with innate immunity. OsEREBP1 interacts with the XA21-binding protein OsXB22a, and overexpression of *OsEREBP1* increases rice immunity against *Xoo* [19]. OsEREBP1 suppresses the negative regulator of bacterial blight resistance, OsNAC2/ONAC004, in the nucleus, while concurrently interacting with OsXB22a to elevate the stability of OsXB22a in the cytoplasm, resulting in increased rice immunity against *Xoo* [20]. Overexpression of *OsERF83* in rice also leads to increased transcription of *PR* genes and increases resistance to *M. oryzae* [21]. OsERF101/OsRAP2.6 interacting with OsRACK1A, OsMPK3, and OsMPK6 increases rice immunity against the compatible *M. oryzae* strain [22]. The activation of *OsERF101* promoter by the rice NAC TF ONAC096 increases immune response to both *M. oryza*e and *Xoo* [23]. OsERF101 is involved in *Xa1*-mediated rice immunity against *Xoo* [24]. OsERF101 interacts with the NLR protein XA1 and transcription activator-like (TAL) effectors (AvrXa7 and Xoo1132), suggesting its formation of a complex with XA1 and the TAL effectors in the nucleus [24]. Both the overexpression and knockout of *OsERF101* increases resistance to *Xoo*, potentially through bHLH and MYB transcriptional regulation pathways, respectively [24].

OsERF922/OsERF92 and OsERF123 act as negative modulators of rice immunity. OsERF922 reduces the expression of defense genes, such as *OsPR1a*, *OsPR1b*, and *OsPR10*, while concurrently elevating abscisic acid (ABA) levels, reducing resistance against *M. oryzae* [25]. CRISPR/Cas9-mediated mutation of *OsERF922* increases resistance to *M. oryzae* and *Xoo* without affecting agronomic traits by activating SA- and JA-dependent immune responses [26,27]. Through whole-genome sequencing and comparative analysis, *OsERF123* was identified as a novel susceptibility gene targeted by the TAL effector TalB [28]. The TalB-mediated promoter activation of *OsERF123* increased susceptibility to *Xoo*.

3. bHLH transcription factor in rice immunity

A bHLH TF contains a conserved bHLH domain comprising approximately 60 amino acids (Fig. 2A). The basic region, spanning approximately 15 amino acids, determines DNA binding specificity, while the helix-loop-helix region functions in protein interactions, such as the formation of homomeric complexes or bHLH heterodimers. Members of the bHLH superfamily bind predominantly to the core sequence CANNTG (E-box), with common variations in plants being CACGTG (G-box) and GATGTG [29].

In rice, over 150 bHLH TFs have been identified, with their domain structures and subfamily classifications previously described [30]. Several of them function as regulators in the immune response to a variety of pathogens (Table S2). The most well-known bHLH TF is MYELOCYTOMATOSIS 2 (MYC2)/OsbHLH9 which is a regulator of JA signaling. Recent studies have demonstrated the role of MYC2 in rice immunity (Fig. 2B). OsMYC2, directly repressed by various OsJAZs through binding to the JAZ-interacting domain, increases resistance to *Xoo* by activating JA signaling [31]. OsMYC2 modulates the expression of genes involved in JA-dependent immunity [32]. Under phosphate-starvation conditions, OsPHR2 activates the *OsMYC2* promoter, leading to increased JA-mediated immunity against *Xoo* [33]. Overexpression of *OsMYC2* increases innate immunity against *M. oryzae* and mitigates the negative effect of ambient temperature (22 °C) on JA-dependent immunity against *M. oryzae* [34]. OsMYC2 also functions in immune response to viruses. OsMYC2-induced JA signaling is

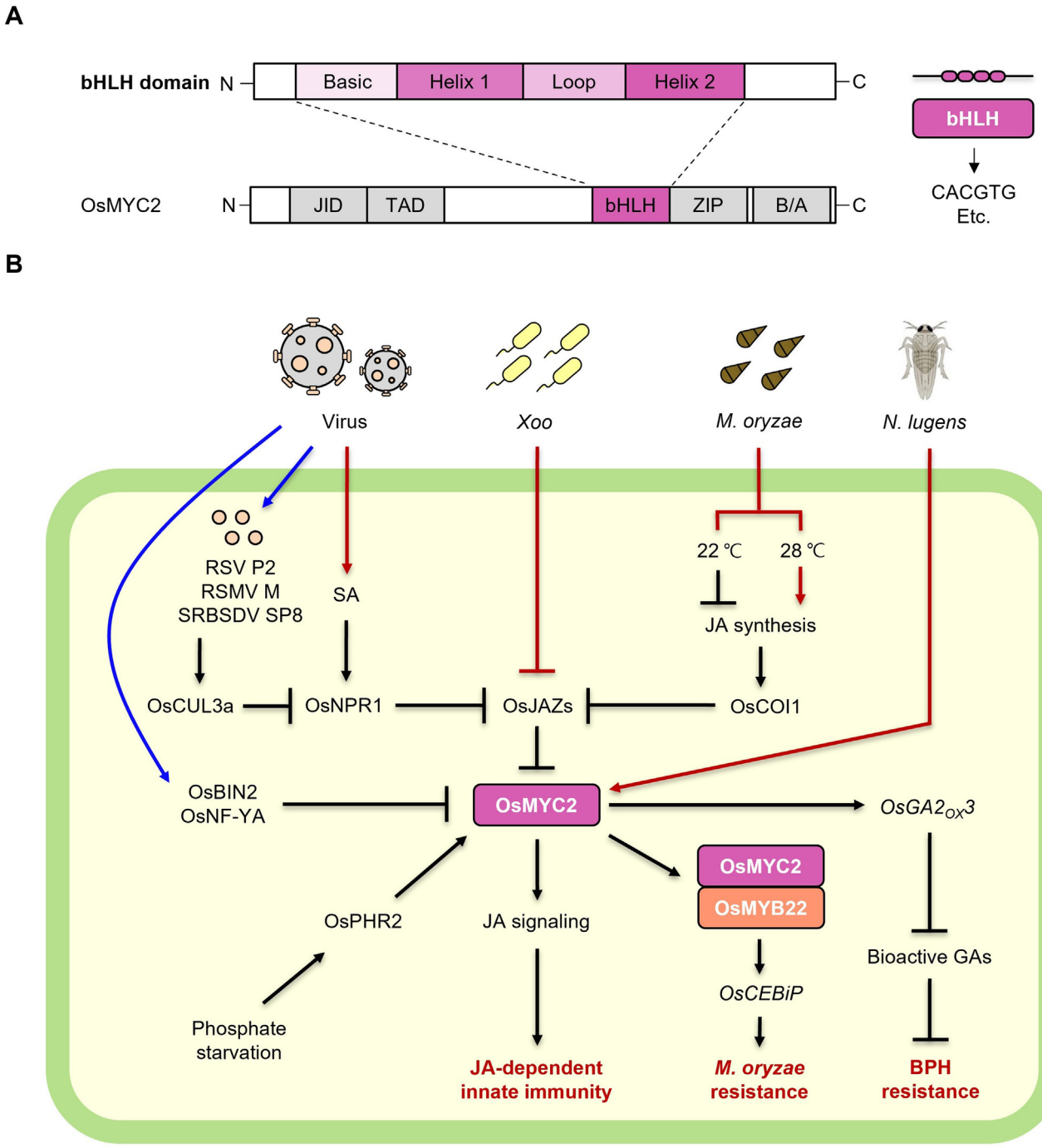


Fig. 2. Central role of OsMYC2 in JA-dependent immunity. (A) Structural domains of the bHLH transcription factor OsMYC2. OsMYC2 contains conserved domains typical of MYC2 proteins, including the JAZ interaction domain (JID), transcription activation domain (TAD), basic helix-loop-helix domain (bHLH), leucine zipper domain (ZIP), and BIF/ACT-like domain (B/A). (B) Schematic illustration of the functions and signaling pathways of OsMYC2 in rice immunity. OsMYC2 activates JA-dependent immunity against various pathogens by regulating the expression of JA- and defense-related genes, such as *OsPRs*, *OsCEBiP*, *OsCsIF6*, and $OsGA2_{OX}3$. Thus, OsMYC2 activation is crucial for innate immunity. OsCOI1, OsNPR1, and OsPHR2 activate OsMYC2 to trigger JA-dependent immunity, while OsJAZ, OsBIN2, OsNF-YA, and OsCUL3a inactivate it. Viral proteins, including RSV P2, RSMV M, and SRBSDV SP8, upregulate the E3 ligase OsCUL3a to degrade OsNPR1, thereby inhibiting the OsMYC2-mediated antiviral response. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility.

essential for brassinosteroid-mediated resistance to rice stripe virus (RSV) [35]. To evade this defense mechanism, RSV upregulates OsBIN2/OsGSK2 to degrade OsMYC2 through phosphorylation [35]. Plant viruses, including RSV, rice black-streaked dwarf virus (SRBSDV), and rice stripe mosaic virus (RSMV), employ viral proteins (RSV P2, SRBSDV SP8, and RSMV M) to impair the transcriptional activity of the OsMYC2 homolog OsMYC3/bHLH10, thus increasing susceptibility by inactivating JA signaling [36]. RSV and SRBSDV also induce the expression of *OsNF-YA* genes, inhibiting OsMYC2 and OsMYC3 to increase susceptibility [37]. The synergistic crosstalk between SA and JA contributes to broad-spectrum antiviral immunity [38]. OsNPR1/OsNH1, a master regulator of SA signaling, interacts with OsMYC2, OsMYC3, and OsJAZs, thereby activating JA signaling by disrupting the OsMYC-OsJAZ complex, resulting in increased resistance to RSV, SRBSDV, and RSMV [38]. However, viral proteins such as RSV P2, SRBSDV SP8, and RSMV

M inhibit the synergistic immune response of SA and JA by promoting OsCUL3a-mediated OsNPR1 degradation and/or dissociating the OsNPR1-OsMYC complex [38]. Furthermore, OsMYC2 increases rice immunity against herbivore attacks. *N. lugens*-activated OsMYC2 binds to two G-box elements in the promoter of the cellulose synthase gene, *OsCslF6*, and upregulates its expression [39]. JA signaling-induced OsCsIF6 synthesizes β-1,3;1,4-d-glucans, and the recognition of the glucan-derived oligosaccharides by the lectin receptor kinase OsLecRK triggers the defense signaling pathway, conferring resistance to BPH [39]. Additionally, OsMYC2 functions in the growth-defense tradeoff during rice-BPH interaction by activating the promoter of $OsGA2_{OX}3$, converting bioactive GAs to inactive forms [40].

Four OsbHLH TFs (i.e., OsbHLH59, OsbHLH61, OsbHLH65, and OsbHLH96) are involved in resistance to BPH. The E-box *cis*-element binding TF OsbHLH65, which does not bind to the G-box, was identified as a binding protein of OsMPK3 using yeast-two hybrid assay [41]. OsMPK3 phosphorylates OsbHLH65 in the nucleus, and the expressions of *OsMPK3* and *OsbHLH65* are induced by biotic stresses (e.g., *N. lugens* and *M. oryzae*) and defense-related phytohormones (MeJA, SA, and ethephon), suggesting that OsbHLH65 functions in the MAPK cascade for rice immunity [41]. OsbHLH59 is also implicated in rice immunity against *N. lugens* as it directly activates the promoter of *OsXIP*, encoding xylanase inhibitor protein, in response to BPH [16]. However, the specific roles of OsbHLH65 and OsbHLH59 in resistance to biotic stress remain unclear. The atypical bHLH transcription factor OsbHLH61, lacking a basic DNA-binding region, interacts with OsbHLH96 and increases resistance to BPH [42]. Expression of some *PR* genes involved in SA-dependent defense response is reduced in *OsbHLH61*-silenced and *OsbHLH96*-expressing plants, indicating that OsbHLH61 is positive regulator of SA-mediated resistance to BPH, and OsbHLH96 negatively regulates this resistance [42]. Additionally, it was proposed [42] that the OsHLH61-OsbHLH96 complex mediates crosstalk between SA and JA signaling pathways, as OsbHLH96 interacts with the JA signaling negative regulator OsJAZ3.

DPF/OsbHLH25, RAI1/OsbHLH3, and OsbHLH6 are regulators of blast resistance. The induction of *DPF*, encoding the OsbHLH TF, by *M. oryzae* leads to elevated production of the antimicrobial compound diterpenoid phytoalexins, momilactones and phytocassanes, coinciding with heightened expression of defense-related genes [43]. RAI1 participates in the PTI signaling pathway and contributes to resistance against *M. oryzae*. Sensing of chitin by OsCERK1 initiates the activation of the RAC/OsRAC1-OsMKK4-OsMPKs (OsMPK3 and OsMPK6) module [44–46], which subsequently phosphorylates RAI1, thereby inducing the expression of defense-related genes such as *OsWRKY19* and *OsPAL1* encoding the initial enzyme of the phenylpropanoid pathway [46]. However, *M. oryzae*-induced OsbHLH6, potentially mediated via TAL effectors, increases JA signaling by activating OsMYC2 and suppresses SA signaling by inactivating OsNPR1, thereby increasing susceptibility to *M. oryzae* [47]. To counteract this, an upregulated OsNPR1 proteins translocate OsbHLH6 from the nucleus to the cytoplasm via direct interaction, leading to increased resistance against *M. oryzae* through activating SA signaling [47]. The study demonstrates that the interaction between OsNPR1 and OsbHLH6 in rice modulates the antagonistic relationship between SA and JA pathways, supporting that cytoplasmic NPR1 is crucial for this crosstalk.

OsbHLH TFs function in bacterial blight disease resistance. OsbHLH57 was identified as a binding protein to the AATCA *cis*-element responsive to *Rhizoctonia solani* and *Xoo* [48]. Functional study demonstrated that OsbHLH57 increases BSDR to *R. solani* and *Xoo* by regulating the expression of various defense- and stress-related genes containing the AATCA element [48]. OsbHLH34, induced by JA, interacts with OsJAZ9, and its overexpression induces transcription of multiple JA-responsive defense genes, increases JA sensitivity, and elevates lignin content [49]. A recent investigation highlighted the importance of the JA-activated MYC2-MYB46 module in lignin biosynthesis in Arabidopsis (*Arabidopsis thaliana*) [50]. These findings suggest that OsbHLH34 increases resistance to *Xoo* through JA-mediated lignin biosynthesis. Moreover, transcriptome analysis revealed various *OsbHLH* genes that respond to compatible and/or incompatible *Xoo* strains [51].

4. bZIP transcription factor in rice immunity

bZIP TFs are characterized by the presence of the conserved bZIP domain, consisting of 60–80 amino acids (Fig. 3A). The highly conserved basic region, featuring the $N(X)_7R/K$ motif, determines DNA binding specificity, while the variable leucine zipper region, characterized by the $L(X)_6L(X)_6L$ motif, is crucial for homo- or heterodimerization conferring transcriptional activity. bZIP TFs can be classified into 10 groups (A–I and S) based on structural features, with those in group D known as regulators in plant immunity or development [52]. While bZIP TFs involved in abiotic stress bind primarily to ACGT-containing ABA-response elements (ABREs) such as TACGTA (A-box), GACGTC (C-box), and G-box, those involved in biotic stress predominantly bind to *cis*-elements with the core sequence TGACG (TGA element) [53,54]. In rice, the genome encodes approximately 90 bZIP transcription factors, with 25 conserved motifs identified outside the bZIP domain [55]. The functions of several of these transcription factors involved in immune responses have been elucidated as detailed below (Fig. 3B; Table S3).

The bZIP TF TGAs interact with NPRs and function in the defense response to pathogen attacks by upregulating defense-related genes, including *PR* genes [4,56]. Among the five rice NHs, OsNPR1 and OsNH3 increase resistance to biotic stresses [57–59]. They strongly interact with most OsTGAs, including rTGA2.1/OsbZIP63, rTGA2.2/OsTGA3/OsbZIP28, and rTGA2.3/OsTGA5/OsbZIP3 [60,61]. Genetic studies have demonstrated that rTGA2.1 and rTGA2.3 act as negative regulators of rice immunity. Transgenic plants expressing dominant-negative *rTGA2.1* and *rTGA2.1*-silenced plants show increased rice immunity against *Xoo* [62]. rTGA2.3 reduces disease resistance to *M. oryzae* by downregulating defense-related genes, suppressing SA accumulation, and attenuating PTI [63]. However, phosphorylation of rTGA2.3 by OsCK2 inhibits its binding to the promoters of *OsPR* genes, resulting in increased immunity to *M. oryzae* [63]. On the other hand, OsTGA2/OsbZIP8, interacting with rTGA2.2, rTGA2.3, and OsNPR1, increases resistance to *Xoo* by upregulating defense-related genes responsive to SA [64].

Non-TGA bZIP TFs also contribute to rice immunity. RF2a/OsbZIP75 and RF2b/OsbZIP30, involved in rice development, regulate the promoter activity of rice tungro bacilliform virus (RTBV) by directly binding to the Box II element in the promoter region [65,66]. These TFs increase disease resistance against RTBV [67]. Overexpression of *RF2a* elevates defense response to *Xoo* and upregulates the transcription levels of defense-related genes, such as *OsPR10a* [68]. APIP5/OsbZIP53 was one of the 12 binding proteins with the *M. oryzae* effector AvrPiz-t [69]. Silencing of *APIP5* reduces the expression of cell death negative regulator genes (e.g., *OsMT2b* and *HLM1*) and induces defense-related genes (e.g., *CHT1* and *CHT3*), resulting in severe cell death and increased innate immunity against *M. oryzae* [70]. Under normal growth condition, APIP5 functions primarily as an RNA-binding protein in the cytoplasm, promoting the turnover of defense-related mRNAs such as *OsLSD1* and *OsRac1* [71]. However, upon infection with *M. oryzae*,

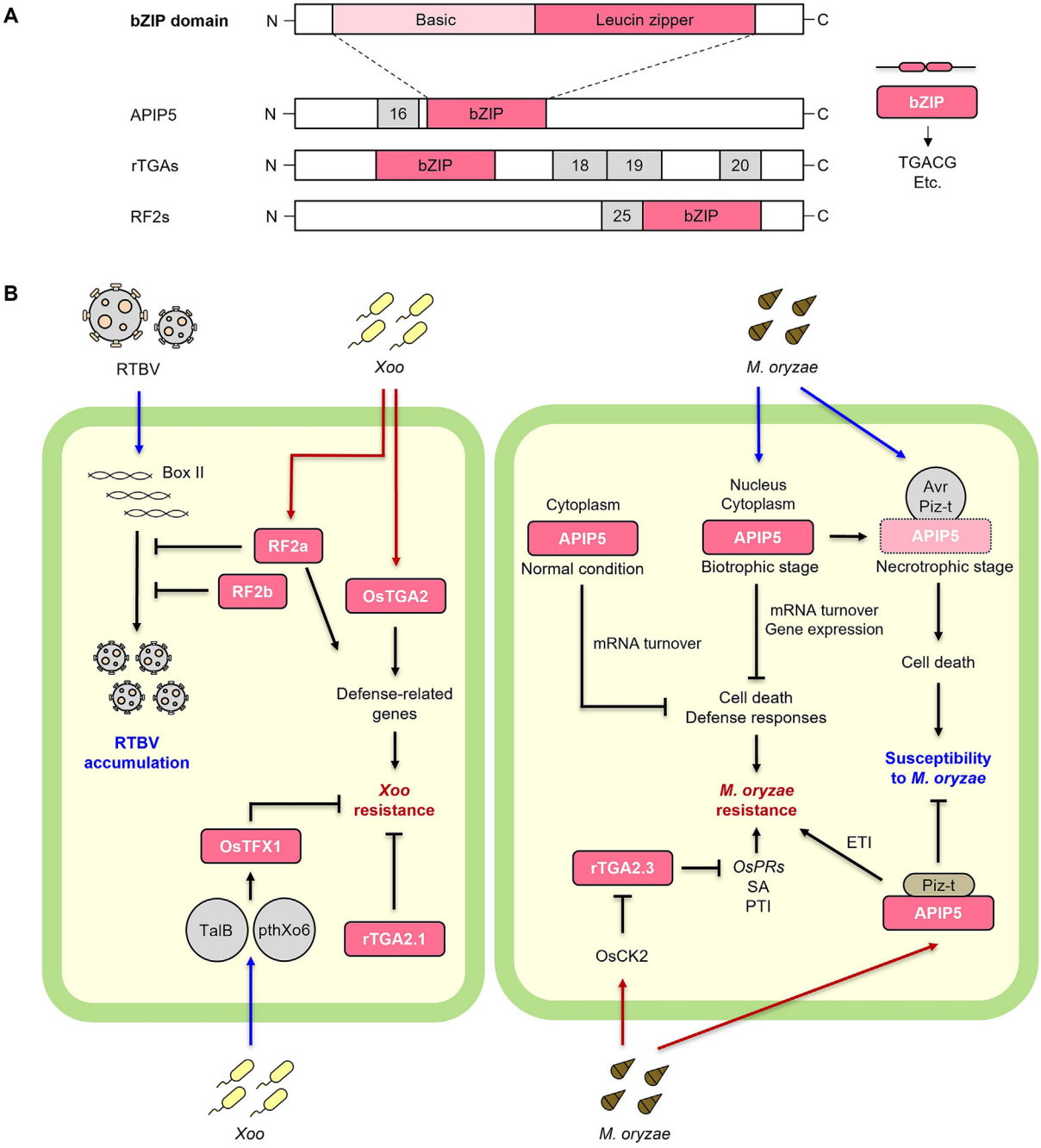


Fig. 3. Functions of bZIP transcription factors in rice-pathogen interactions. (A) Structural domains of OsbZIP transcription factors. The bZIP superfamily is characterized by the presence of the bZIP domain. bZIPs involved in biotic stress predominantly bind to the TGA element (TGACG). OsbZIP proteins can be classified based on the presence of 25 conserved additional motifs [55]. APIP5 includes motif 16; rTGA2.1, rTGA2.3, and OsTGA2 include motifs 18–20; and RF2a and RF2b include motif 25. (B) Schematic illustration of the functions of OsbZIPs in biotic stress responses. RF2a, RF2b, and OsTGA2 repress RTBV accumulation and/or susceptibility to *Xoo*, while OsTFX1, rTGA2.1, and rTGA2.3 suppress resistance to *Xoo* or *M. oryzae*. APIP5 also suppresses innate immunity to the hemibiotrophic pathogen *M. oryzae* by inhibiting cell death and defense responses. However, during the necrotrophic stage, the AvrPiz-t effector binds to and leads to the degradation of APIP5, increasing susceptibility to *M. oryzae* by inducing cell death. To counteract this, Piz-t competes with AvrPiz-t to interact with this bZIP, stabilizing both APIP5 and Piz-t and triggering ETI. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility.

APIP5 localizes to both the nucleus and cytoplasm, where it inhibits the expression of defense-related genes and increases mRNA turnover, respectively, thus favoring the biotrophic stage of the pathogen [71]. For the necrotrophic stage of *M. oryzae*, the effector AvrPiz-t promotes cell death by downregulating the transactivation activity and protein level of APIP5 [70]. To counteract this, the NLR R protein Piz-t competes with AvrPiz-t to bind APIP5, stabilizing both APIP5 and Piz-t and triggering ETI [70]. OsTFX1/OsbZIP73 reduces defense response to *Xoo*. The induction of *OsTFX1* by the TAL effector pthXo6 or TalB of the *Xoo* strains is necessary for full virulence, indicating *OsTFX1* as a susceptibility gene for *Xoo* infection [28,72].

OsbZIP50 and OsbZIP81 are also suggested to contribute to innate immunity. OsbZIP50, referred to as OsbZIP60 in the referenced article, was identified as a binding partner of Bphi008a conferring resistance to BPH using yeast-two hybrid assay [73], and it

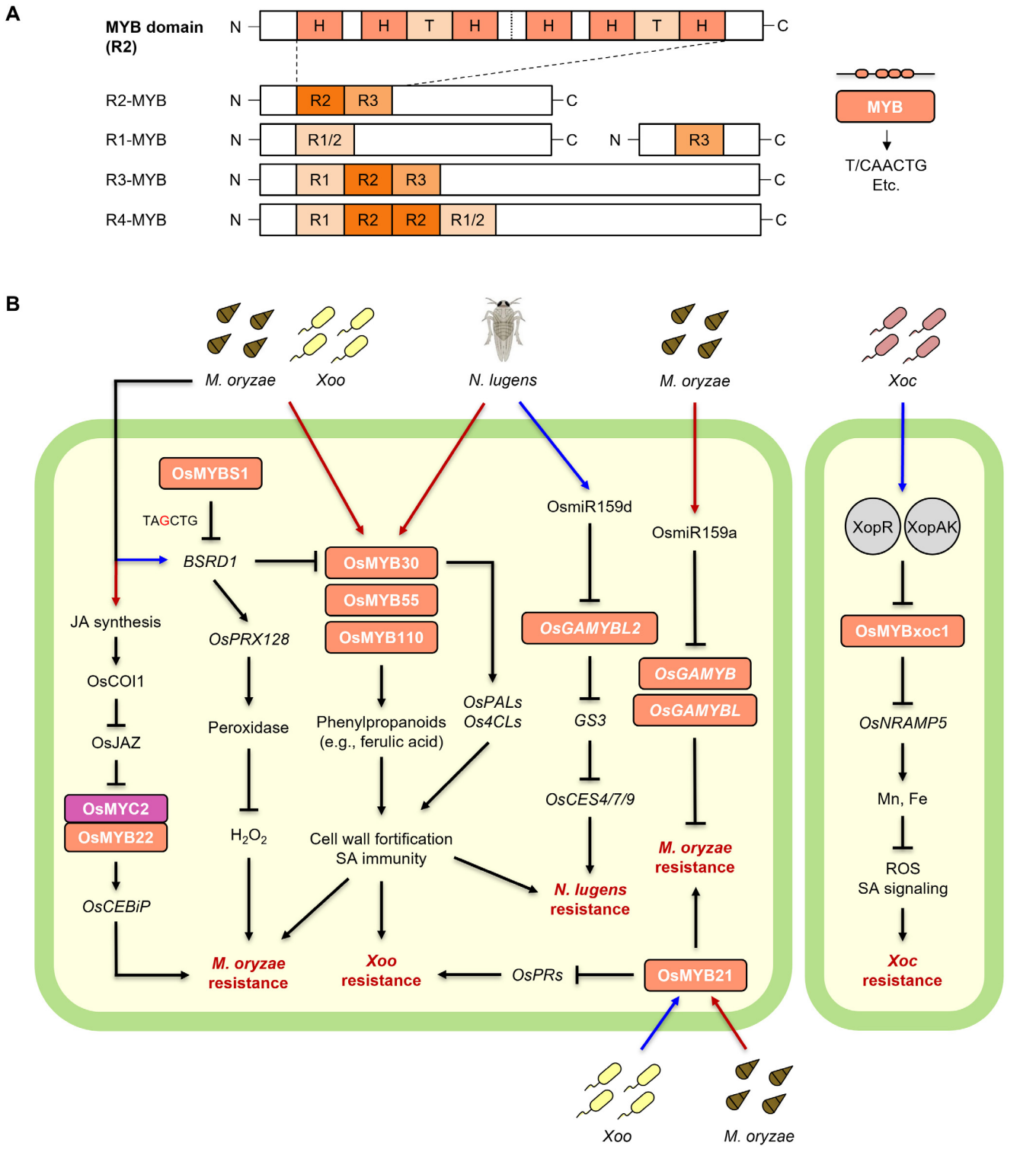


Fig. 4. The OsMYB superfamily-mediated regulatory network of immune responses in rice. (A) Structural features of OsMYB subfamilies. The OsMYB superfamily is categorized into four subfamilies based on the number of MYB domains, which are divided by the number of incomplete repeats they contain (R1–R4): 1R-, 2R-, 3R-, and 4R-MYB. MYB transcription factors bind diverse *cis*-elements, such as the MBS (T/CAACTG) motif, to regulate various plant processes. (B) Schematic illustration of the regulatory roles of the OsMYBs in rice immunity. OsMYBS1 binds strongly to the single-nucleotide polymorphisms TAGCTG sequence of the *BSRD1* promoter and represses its expression, resulting in increased resistance to *M. oryzae*. Other MYBs including OsMYB22, OsMYB30, OsMYB55, OsMYB110, OsGAMYBL2, and OsMYBxoc1 act as positive regulators of rice immunity, whereas OsGAMYB and OsGAMYBL negatively regulate it. Overexpression of *OsMYB21* increases blast resistance but reduces resistance to bacterial blight. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility.

was also predicted to be involved in OsNRP1-mediated SA immunity against *M. oryzae* [74]. Liu et al. demonstrated that OsbZIP81 and OsbZIP84, rice homologs of Arabidopsis VIP1, are typical bZIP TFs predominantly binding to the *cis*-element OVRE (core sequence GCTG), and proposed that OsbZIP81 increases defense response by regulating JA signaling and metabolism pathways [75].

5. MYB transcription factor in rice immunity

The MYB gene family, initially discovered in the avian myeloblastosis virus, is ubiquitous in eukaryotic organisms. MYB TFs feature a DNA-binding domain comprising 1–4 incomplete repeats (R1–R4), with each repeat consists of approximately 50 amino acids that form a helix-turn-helix (HTH) structure. MYB TFs can be categorized into four subfamilies: 1R-, 2R-, 3R-, and 4R-MYB, according to the number of MYB domains they contain (Fig. 4A). The R2R3 MYB TFs, belonging to the 2R-MYB subfamily, are the most prevalent MYB proteins in plants and function in a multitude of plant processes, encompassing reactions to biotic stressors [76]. MYB TFs bind to diverse *cis*-elements including MBS (T/CAACTG), MRE (CAACCA and ACCTAA), MSA (AACGG), and MBSII (TAACTAAC) [77–79]. A recent study identified 190 *OsMYB* genes in the rice genome, with 99 genes encoding R2R3-MYB TFs [80]. Despite their considerable numbers, only a limited OsMYB TFs have been functionally determined in rice immunity (Fig. 4B; Table S4).

Interplays between OsMYBs and BSRD1 have been extensively elucidated. OsMYBS1, recognized as a TATCCA element-binding TF involved in sugar signaling, serves as a critical regulator of innate immunity against *M. oryzae*. BSRD1 increases susceptibility to *M. oryzae* races by directly upregulating the expression of the peroxidase gene *OsPRX128*, which degrades H_2O_2 synthesized by PTI [81]. However, in the rice blast resistance variety Digu, OsMYBS1 strongly binds to the single nucleotide changed *BSRD1* promoter, repressing its expression and increasing resistance to *M. oryzae* races through H_2O_2 accumulation [81]. The PAMP-activated OsMKK4-OsMPKs-OsMYBs (i.e., OsMYB30, OsMYB55, and OsMYB110) signaling pathway selectively induces the expression of genes involved in the shikimate and cinnamate/monolignol pathways by binding to CA/CCCAA/CCCA/CC (AC-rich element) [82]. Consequently, three OsMYBs increase disease resistance to the blast pathogen *Pyricularia oryzae* as well as *Xoo* by promoting the biosynthesis of ferulic acid, which strengthens the cell wall [82]. OsMYB30 directly activates the promoters of *OsPAL6* and *OsPAL8*, which contain AC-rich elements, to increase lignin and SA contents in response to BPH infestation, thereby increasing resistance to the insect pest [83]. OsMYB30 is linked with the OsMYBS10-BSRD1 module. Due to the indirect repression of *OsMYB30* expression by BSRD1, the OsMYBS1-mediated suppression of *BSRD1* in Digu increases the transcript levels of *OsMYB30* [84]. OsMYB30 binds to conserved TCCTCCT/GC and TCGC/TCG motif in the promoter of *Os4CL3* and *Os4CL5*, activating their promoter activities and resulting in increased blast resistance through increased lignin production and reinforced cell walls [84].

MicroRNA-mediated suppression of *OsMYB* also functions in rice-pathogen interactions. *M. oryzae*-induced OsmiR159a increases resistance to the pathogen and reduces transcription of the GA-related *MYB* genes *OsGAMYB* and *OsGAMYBL* [85]. Given that mutations of *OsGAMYB* or *OsGAMYBL* resulted in increased blast resistance, these two MYB TFs are negative regulators of innate immunity against *M. oryzae* [85]. However, OsmiR159d negatively regulates the defense response to *N. lugens* by suppressing the transcript level of *OsGAMYBL2*, which confers resistance to *N. lugens* [86]. OsGAMYBL2 binds to the TAACCG element on the promoter of *GS3*, encoding guanine nucleotide-binding protein gamma subunits, and represses its expression [86]. Genetic and molecular evidence showed that GS3 attenuates BPH resistance and down-regulates the expressions of the cellulose synthase genes (i.e., *OsCESA4*, *OsCESA7*, and *OsCESA9*), suggesting that the OsmiR159-OsGAMYBL2-GS3 signaling pathway increases BPH resistance through cellulase synthesis [86].

Recently, two research groups elucidated the biological functions and mechanisms of OsMYBs in innate immunity. Qiu et al. demonstrated the significance of ambient temperature and JA-dependent OsMYB22 in rice blast resistance [34]. Upon *M. oryzae* induction, JA prompts OsMYC2 to interact with OsMYB22, leading the MYB TF to bind to and activate the promoter of defense-related genes, such as *OsCEBiP*, which encodes chitin-binding protein and confers PTI [34]. However, this JA signaling is attenuated at 22 °C, resulting in a decrease in rice immunity against *M. oryzae* compared to the normal growth temperature of 28 °C [34]. Zhang et al. identified OsMYBxoc1/OsMYB67 as a crucial regulator conferring rice immunity against *Xanthomonas oryzae* pv. *oryzicola* (*Xoc*), a major causative agent of bacterial leaf streak [87]. OsMYBxoc1, which is induced by *M. oryzae* and SA but reduced by *Xoc* type III effectors such as XopR and XopAK, regulates the expression of various genes involved in defense responses such as *OsNRAMP5*, which encodes a Mn and Fe metal transporter that inhibits SA and ROS accumulation, thereby increasing blast resistance [87]. However, the XopAK effector directly inactivates the *OsMYBxoc1* promoter, leading to increased OsNRAMP5-mediated metal ion transport activity, inhibiting SA signaling and flag22-induced ROS accumulation, thereby increasing susceptibility to *Xoc* [87].

Other OsMYBs are also implicated in innate immunity. Genome-wide computational and gene expression analyses identified the OsMYB3 binding motif, AA/CCA/TAA/CC, and three potential defense-related target genes of this TF, although OsMYB3 alone was found to be insufficient to regulate the expression of these target genes [88]. Ectopic expression of *OsMYB4* in Arabidopsis and tomato confers increased disease resistance [89,90]. Overexpression of *OsMYB21/OsJAMYB* in rice increases blast resistance [91], but reduces rice immunity against *Xoo* and the expression of various *PR* genes [92].

6. NAC transcription factor in rice immunity

The plant-specific NAC superfamily functions in plant growth, development, and stress responses, including innate immunity. NAC TFs can be classified into typical and atypical proteins. Typical NAC TFs consist of a highly conserved N-terminal region containing the NAC domain, responsible for DNA-binding and oligomerization, and a diverse C-terminal transcriptional regulatory (TR) domain (Fig. 5A). Atypical NAC TFs contain either only the NAC domain or the NAC domain along with another domain, such as a transmembrane domain [93,94]. Although many NAC TFs bind primarily to the canonical NAC recognition sequence (NACRS) element (CATGT or CACG) [95], diverse NAC-binding elements including the iron deficiency responsive cis-acting element 2 (IDE2), JUB1 binding site (JBS), JBS-like (JBSL), the secondary wall NAC-binding element (SNBE), the ACGCAA element, and the Mutator-derived MITE element have been identified [96–103].

In rice, there are approximately 150 NAC TFs, some of which positively or negatively regulate resistance to pathogens (Fig. 5B; Table S5). Nakashima et al. first demonstrated the function of OsNAC in rice-pathogen interactions, showing that *OsNAC6/ONAC048* overexpression enhances innate immunity to *M. oryzae* through upregulating gene expressions of *OsPRs* and *OsPRXs* but leads to yield penalty, indicating the role of OsNAC6 in controlling the growth-defense tradeoff [104]. In response to the incompatible strain of *Acidovorax avenae*, OsNAC4/ONAC068 is induced and translocates into the nucleus in a phosphorylation-dependent manner to regulate the expressions of genes (*OsHSP90* and *IREN*), resulting in hypersensitive cell death accompanied by the loss of plasma membrane integrity, nuclear DNA fragmentation, and typical morphological changes [105]. Silencing of *OsNAC122* or

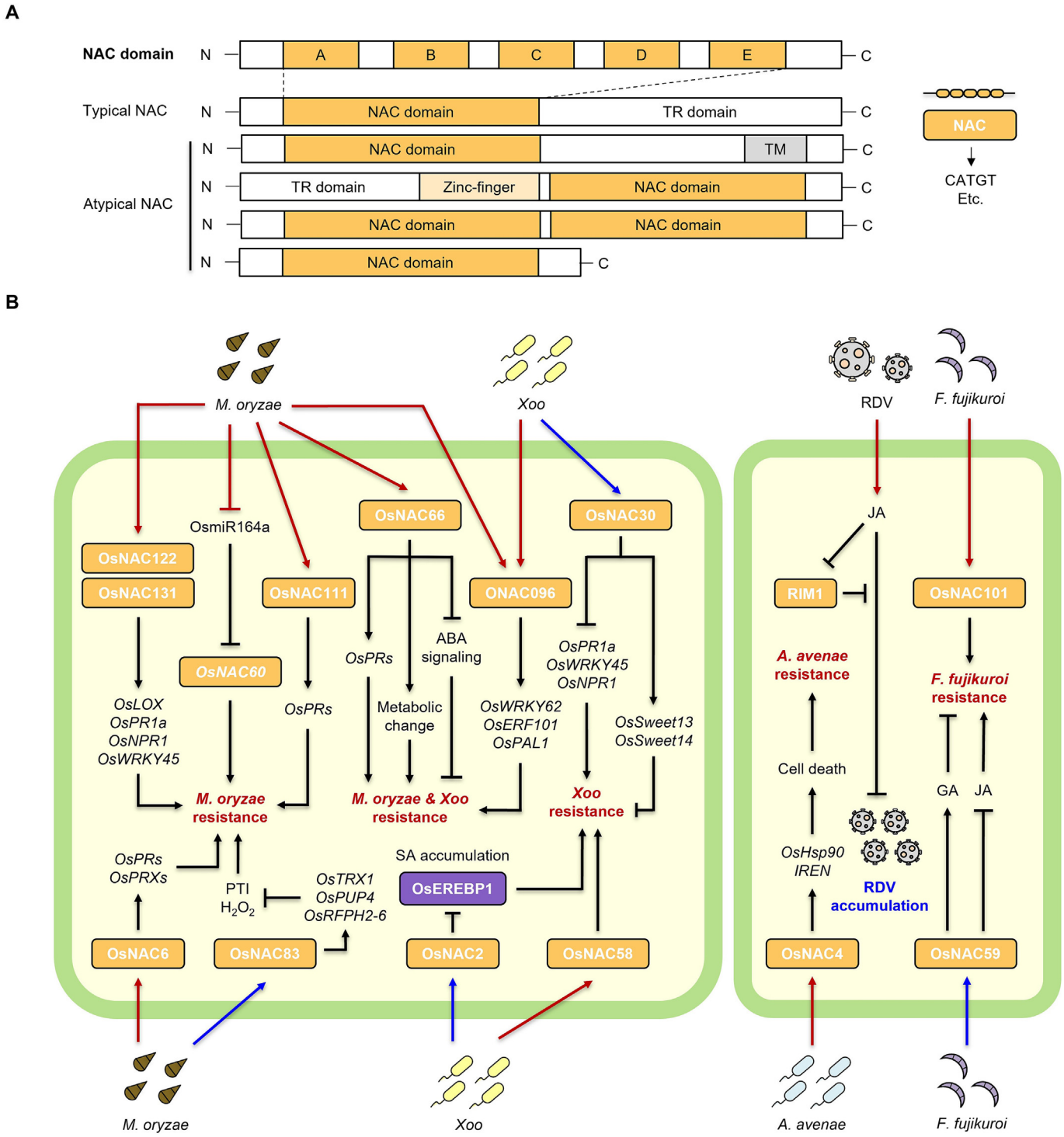


Fig. 5. Diverse functions of the OsNAC superfamily in rice immunity. (A) Structural features of typical NAC transcription factors. NAC transcription factors are composed of a highly conserved N-terminal NAC domain, which contains five conserved subdomains (A to E), and a diverse C-terminal transcriptional regulatory (TR) domain. The NAC superfamily binds to various elements, such as the NAC recognition sequence (CATGT or CACG), to modulate the expression of numerous genes involved in plant growth, development, and stress responses. (B) Schematic illustration of the functions and regulatory mechanisms of OsNACs in defense responses. Ten OsNACs, including OsNAC4/ONAC068, OsNAC6/ONAC048, OsNAC58, OsNAC60, OsNAC66, OsNAC96, OsNAC101, OsNAC111, OsNAC122, and OsNAC131, enhance rice immunity. In contrast, five OsNACs, including RIM1/ONAC054, OsNAC2/ONAC004, OsNAC30, OsNAC59, and OsNAC83, increase susceptibility to pathogens. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility.

OsNAC131 reduces resistance to *M. oryzae* and downregulation of defense-related genes such as *OsHI-LOX*, *OsPR1a*, *OsWRKY45*, and *OsNPR1* [106]. Overexpression of *OsNAC111* increases rice immunity to *M. oryzae* via upregulating various *OsPR* genes [107]. Overexpression of *OsNAC58* in rice also increases resistance to *Xoo* [108]. OsmiR164a-mediated suppression of *OsNAC60* attenuates the defense response to *M. oryzae*, indicating that OsNAC60 is a positive regulator of blast resistance [109]. Most recently, OsNAC TFs involved in rice immunity were identified using the *OsNAC* mutant library [110]. *OsNAC101* mutation decreased resistance to bakanae disease caused by *Fusarium fujikuroi*, indicating OsNAC101 positively regulates rice immunity against the fungal pathogen

[110]. The molecular mechanism of OsNAC101 remains to be elucidated in the future.

OsNAC66 and OsNAC96 contribute BSDR to *M. oryzae* and *Xoo*. Overexpression of *OsNAC66* increases rice immunity against both *M. oryzae* and *Xoo* by inhibiting ABA signaling and promoting the accumulation of sugars and amino acids [111]. OsNAC66 increases blast resistance by directly activating the promoters of *OsWRKY62*/*XB10* and *cytochrome P450* genes encoding enzymes involved in the biosynthesis of diterpenoid phytoalexins, conferring resistance to fungal and bacterial pathogens [112]. Overexpression of *OsNAC96* increases PTI and BSDR to *M. oryzae* and *Xoo*, while loss-of-function results in the opposite effect [23]. OsNAC96 induces the expression of three defense-related genes (i.e., *OsERF101*, *OsWRKY62*, and *OsPAL1*) by activating their promoters, and it interacts with OsNAC66 [23]. However, the effect of the interaction between OsNAC66 and OsNAC96 remains unknown.

Five NAC TFs have been identified as negative regulators of rice immunity. The rice retrotransposon *Tos17* insertional mutant of *RIM1/ONAC054* leads to a reduction in the accumulation of the rice dwarf virus (RDV) capsid protein, while overexpression of *RIM1* increases RDV multiplication and infection symptoms in rice [113]. RIM1, which is targeted for degradation via the 26S proteasome pathway in response to JA, suppresses JA-dependent immunity against RDV [114,115]. Thus, RIM1 functions as a negative regulator of resistance to RDV. OsNAC83 decreases innate immunity against *M. oryzae* by suppressing defense-related genes, H_2O_2 accumulation, and PTI [103]. OsNAC83 binds to the ACGCAA element and activates the promoters of *OsTRX1*, *OsPUP4*, and *OsRFPH2-6* [103]. OsNAC83-induced OsRFPH2-6 represses blast resistance and PTI, indicating that OsNAC83 functions as negative regulator of innate immunity via OsRFPH2-6 [103]. OsNAC2 suppresses SA accumulation by downregulating the expression of genes associated with SA biosynthesis, and interacts with OsEREBP1 to facilitate the degradation of OsEREBP1, consequently leading to increased susceptibility to *Xoo* [20]. Mutagenesis of *OsNAC30* and *OsNAC59* increases resistance to *Xoo* and *F. fujikuroi*, respectively [110]. *Xoo*-induced expressions of SA immunity-related genes (i.e., *OsPR1a*, *OsWRKY45*, and *OsNPR1*) are significantly increased in *osnac30* mutant lines, while transcripts of *OsSWEET13* and *OsSWEET14* are reduced [110]. In *osnac59* mutant lines, GA signaling conferring susceptibility to *F. fujikuroi* is repressed, while JA signaling, important for bakanae resistance, is induced [110]. These results revealed that OsNAC30 and OsNAC59 act as negative regulators of rice immunity against *Xoo* and *F. fujikuroi*, respectively.

7. WRKY transcription factor in rice immunity

WRKY TFs are categorized based on the presence of a WRKY domain (WD), which comprises a highly conserved heptad WRKY motif, WRKYGQK, located in the N-terminus, and a zinc finger domain situated in the C-terminus (Fig. 6A). These TFs are further classified into three groups: Group I TFs possess two WDs, Group II TFs contain one WD and one C_2H_2 zinc finger, and Group III TFs carry one WD and one C_2HC zinc finger [116]. Predominantly, WRKY TFs bind to the W-box motif (TTGACC/T) and the W-box-like element 1 (WLE1; TGACA) [117,118]. However, they are anticipated to bind to a diverse array of sequences within the promoters of target genes associated with various plant processes, including disease resistance [119]. In rice, there are over 120 OsWRKY members, and the functions of many OsWRKYs in rice immunity against various pathogens have been explored (Table 1).

A recent study demonstrates a novel mechanism of translational regulation in OsWRKY7-mediated immunity. OsWRKY7 exhibits binding affinity to the WLE1 motif located on the promoter region of *OsPR10a* and directly instigates its activation [124]. This TF prompts the expression of diverse *PR* genes, thereby bolstering BSDR to *M. oryzae* and *Xoo* [124,125]. Alternative translation mechanism involving the main open reading frame (mORF) and the downstream in-frame ORF (diORF) produces two variants of OsWRKY7 proteins, with differing stabilities [126]. The full-length OsWRKY7, which encompasses the N-terminal 26S proteasome degradation region, exhibits instability, but its stability is increased by flg22 and *Xoo*. However, the short-length OsWRKY7 lacking the N-terminal region remains consistently stable and functional. mORF hinders the initiation of diORF translation; hence, mutations in mORF facilitate the production of short-length OsWRKY7, bolstering rice immunity against *Xoo* by upregulating *PR* gene expressions and ROS synthesis [126]. Zheng et al. showed the potential for alternative translation-mediated regulation of protein stability in other OsWRKY proteins such as OsWRKY10 and OsWRKY26 [126]. Translational control is a strategy for augmenting plant stress resilience [193]. Thus, diORF translation of *WRKYs* emerges as an avenue for increasing crop resilience.

The OsWRKY TFs serve as crucial components of the MAPK transduction pathway in rice immunity (Fig. 6B). Upon *Xoo* infection, the activation of OsMKK3-OsMPK7 signaling pathway triggers the phosphorylation of OsWRKY30 by OsMPK7, subsequently increasing bacterial blight resistance by stimulating the expression of defense-related genes [141]. MAPK signaling modulates the expression of *OsWRKY30*. OsMPK6 directly interacts with and phosphorylates the CCCH-type zinc finger protein OsLIC, which is a negative regulator of rice immunity against both *Xoo* and *Xoc*, thereby alleviating its inhibition on the promoter activity of *OsWRKY30* [142]. These studies show the multifaceted regulations of OsWRKYs by MAPK cascades, elucidating their roles in activating rice immunity. Moreover, OsWRKY45, an important regulator of SA-dependent immunity, is implicated in MAPK signaling pathways. The JA-responsive OsVQ13 directly interacts with OsMPK6 and increases resistance to *Xoo* potentially by inhibiting the interaction between OsMPK6 and phosphatases, thereby maintaining OsMPK6 activity and activating the OsMPK6-OsWRKY45 module [156]. Furthermore, OsMPK6, activated by OsMKK10-2 in response to SA, phosphorylates OsWRKY45, and it results in SA-dependent immunity against *M. oryzae* by increasing the transcriptional activity of OsWRKY45 [149,153]. However, the ABA-activated tyrosine phosphatases such as OsPTP1 and OsPTP2 inactivate OsMPK6 through dephosphorylation, underscoring the interplay between ABA and SA signaling via the OsPTPs-OsMPK6-OsWRKY45 [153]. The OsMKK10-2-OsMPK3-OsWRKY55 cascade also modulates disease resistance and phytohormone crosstalk [167]. The OsREIW1-mediated ubiquitination of OsWRKY55, previously referred to as OsWRKY31, initiates its proteasomal degradation. However, phosphorylation of OsWRKY55 by the PAMP-activated OsMKK10-2-OsMPK3 module enhances its stability and DNA-binding activity. This phosphorylation event results in increased blast resistance and reduced plant growth via the upregulation of SA and JA pathways, and the downregulation of IAA pathway. OsWRKY53 exhibits a dual regulatory role in plant defense, positively modulating resistance to *M. oryzae* and *N. lugens* while reducing resistance to *C. suppressalis* and *Xoo* [161–165]. The fungal PAMP-responsive OsMKK4-OsMPK1/3/6 module phosphorylates OsWRKY53, which is also rapidly induced by chitin and *M. oryzae*, and increases its transcriptional activity, thereby increasing blast resistance by the induction of defense-related genes [161,162]. In the context of insect pests, OsWRKY53 increases resistance to SSB while decreasing resistance to BPH. OsMPK3 contributes to resistance against *C. suppressalis* by upregulating JA-induced trypsin protease inhibitors [194]. However, OsMPK3/6-phosphorylated OsWRKY53 acts as a negative regulator of OsMPK3/6 activities, JA, ET, and trypsin inhibitors, thereby resulting in reduced defense against *C. suppressalis*

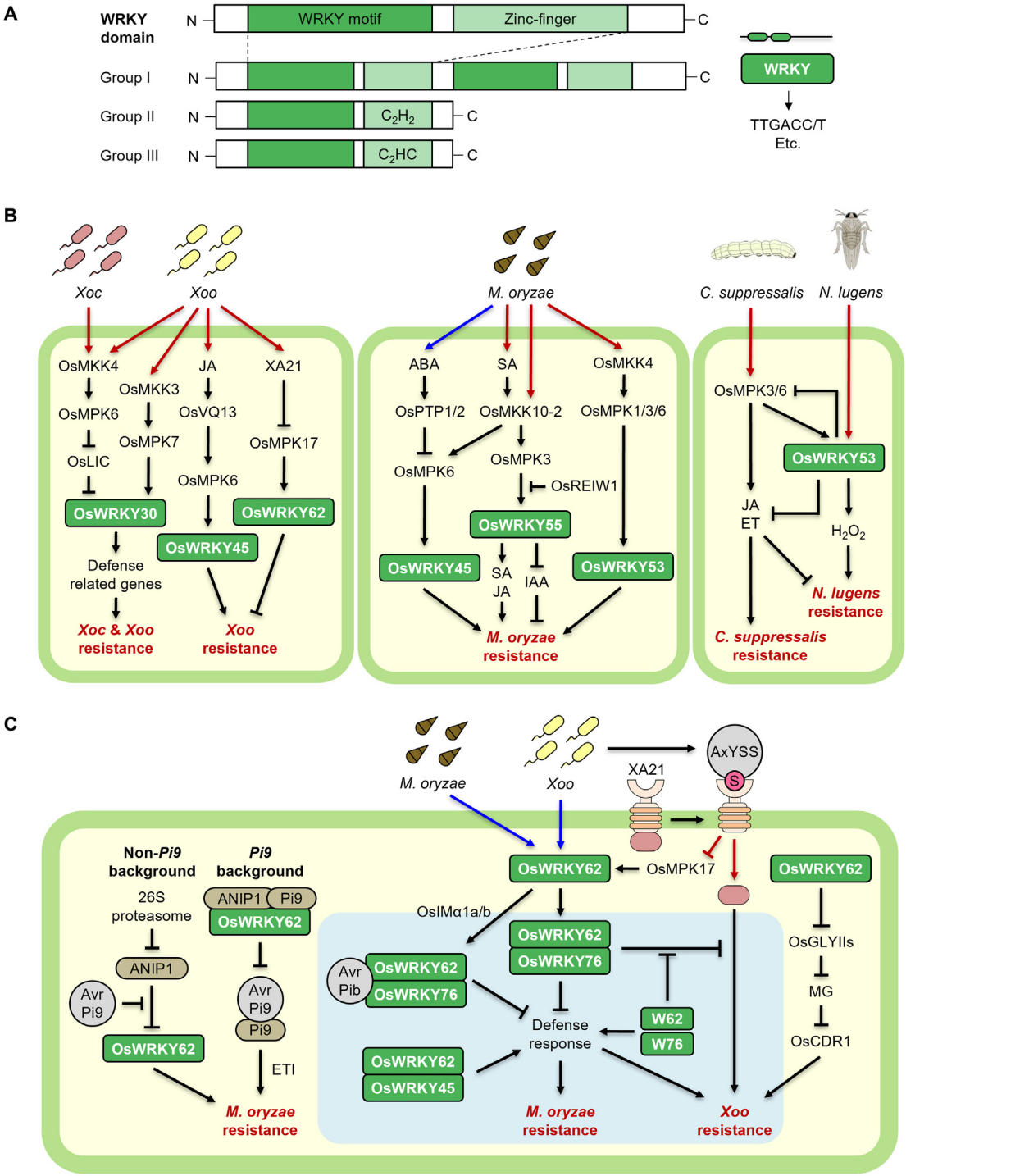


Fig. 6. The regulatory mechanisms and signaling pathways of OsWRKY transcription factors in immune responses. (A) Structural features of WRKY groups. The WRKY superfamily is classified into three groups based on the number of WRKY domains and the types of zinc-finger domain. WRKY transcription factors primarily binding to the W-box motif (TTGACC/T) and the WLE1 motif (TGACA). (B) Schematic illustration of the OsMPK-OsWRKY signaling pathway conferring rice immunity. The modules including OsMKK4-OsMPK6-OsLIC-OsWRKY30, OsMKK3-OsMPK7-OsWRKY30, OsVQ13-OsMPK6-OsWRKY45, OsMKK10-2-OsMPK6-OsWRKY45, and OsMKK10-2-OsMPK3-OsWRKY55 contribute to increased rice immunity, while the OsMPK17-OsWRKY62 module attenuates it. OsWRKY53, regulated by OsMPKs, positively influences resistance against *M. oryzae* and *N. lugens*, but negatively modulates resistance against *C. suppressalis*. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility. (C) The distinct roles and regulatory mechanisms of OsWRKY62 in rice immunity. Full-length OsWRKY62 and OsWRKY76 form homo- and hetero-oligomers that repress rice immunity against *M. oryzae* and *Xoo* by inducing metabolic and transcriptional alterations. Specifically, full-length OsWRKY62 directly represses OsGLY II glyoxalases, thereby inhibiting OsCDR1-mediated immunity against *Xoo* through the accumulation of the sugar metabolite methylglyoxal (MG). Conversely, the truncated forms (W62 and W76) increase rice immune responses to these pathogens. OsMPK17 inhibits bacterial blight resistance by upregulating OsWRKY62, and this module is inhibited by *Xoo* infection in *Xa21* background. The OsWRKY62 complex interacts with OsIMα1a and OsIMα1b for nuclear localization, resulting in reduced blast resistance. However, OsWRKY62 increases resistance by forming a heterodimer with OsWRKY45. Additionally, the function of OsWRKY62 as a regulator of blast resistance depends on the presence of *Pi9*. In non-*Pi9* backgrounds, OsWRKY62 enhances immunity, but AvrPi9 stabilizes ANIP1, leading to OsWRKY62 degradation. In *Pi9* backgrounds, OsWRKY62 suppresses resistance by forming a complex with Pi9 and ANIP1. However, this complex is dissociated upon AvrPi9 recognition to activate ETI. Red arrows indicate pathogen-induced defense response. Blue arrows indicate pathogen-induced susceptibility.

Table 1
OsWRKY transcription factors involved in rice-pathogen interactions.

Gene	MSU ID	Pathogen	Reference
OsWRKY4	LOC_Os03g55164	*R. solani* (P)	Wang et al. [120], Peng et al. [121]
OsWRKY6	LOC_Os03g58420	*Xoo* (P)	Choi et al. [118], Im et al. [122], Im et al. [123]
OsWRKY7	LOC_Os05g46020	*M. oryzea* (P), *Xoo* (P)	Tun et al. [124], Son et al. [125], Zheng et al. [126]
OsWRKY10	LOC_Os01g09100	*M. oryzea* (P), *Xoo* (P)	Choi et al. [127], Wang et al. [128]
OsWRKY11	LOC_Os01g43650	*Xoo* (P)	Lee et al. [129]
OsWRKY13	LOC_Os01g54600	*M. oryzae* (P), *Xoo* (P), RDV (P)	Qiu et al. [130], Xiao et al. [131], Cheng et al. [132], Qin et al. [133]
OsWRKY19	LOC_Os05g49620	*M. oryzae* (P), *Xoo* (P)	Du et al. [134]
OsWRKY22	LOC_Os01g60490	*M. oryzae* (P)	Abbruscato et al. [135]
OsWRKY24	LOC_Os01g61080	*M. oryzae* (P)	Yokotani et al. [136]
OsWRKY28	LOC_Os06g44010	*M. oryzae* (N)	Chujo et al. [137]
OsWRKY30	LOC_Os08g38990	*M. oryzae* (P), *R. solani* (P), *Xoc* (P), *Xoo* (P)	Peng et al. [138], Lee et al. [139], Han et al. [140], Jalmi et al. [141], Wang et al. [142]
OsWRKY42	LOC_Os02g26430	*M. oryzae* (N), *Xoo* (P)	Cheng et al. [132], Wang et al. [143]
OsWRKY45	LOC_Os05g25770	*M. oryzea* (P), *N. lugens* (N), *R. solani* (N), *Xoc* (PN), *Xoo* (PN)	Cheng et al. [132], Shimono et al. [144], Tao et al. [145], Shimono et al. [146], Matsushita et al. [147], Jiang et al. [148], Ueno et al. [149], Inoue et al. [150], Nakayama et al. [151], Goto et al. [152], Ueno et al. [153], Huangfu et al. [154], Fukushima et al. [155], Uji et al. [156], Wu et al. [157]
OsWRKY47	LOC_Os07g48260	*Xoo* (P), *M. oryzea* (P)	Choi et al. [127], Wang et al. [128]
OsWRKY51	LOC_Os04g21950	*Xoo* (P)	Choi et al. [127], Hwang et al. [159]
OsWRKY53	LOC_Os05g27730	*C. suppressalis* (N), *M. oryzae* (P), *N. lugens* (P), *Xoo* (N)	Chujo et al. [160], Chujo et al. [161], Yoo et al. [162], Hu et al. [163], Hu et al. [164], Xie et al. [165]
OsWRKY55	LOC_Os03g20550	*M. oryzae* (P)	Zhang et al. [166], Wang et al. [167]
OsWRKY62	LOC_Os09g25070	*M. oryzae* (PN), *Xoo* (PN)	Fukushima et al. [155], Peng et al. [168], Park et al. [169], Zhu et al. [170], Shi et al. [171], Fu et al. [172]
OsWRKY65	LOC_Os12g02470	*F. fujikuroi* (P), *Xoo* (P)	Son et al. [177]
OsWRKY67	LOC_Os05g09020	*M. oryzae* (P), *R. solani* (N), *Xoo* (P)	Vo et al. [178], Liu et al. [179], Zhao et al. [180]
OsWRKY68	LOC_Os04g51560	*Xoo* (P)	Zhu et al. [181]
OsWRKY71	LOC_Os02g08440	*N. lugens* (P), *Xoo* (P)	Liu et al. [182], Chujo et al. [183], Li et al. [184]
OsWRKY72	LOC_Os11g29870	*Xoo* (N)	Hou et al. [185]
OsWRKY76	LOC_Os09g25060	*M. oryzae* (PN), *Xoo* (PN)	Seo et al. [173], Liu et al. [174], Liang et al. [175], Xu et al. [176], Yokotani et al. [186]
OsWRKY80	LOC_Os03g63810	*R. solani* (P)	Peng et al. [121]
OsWRKY88	LOC_Os07g40570	*Xoo* (P)	Choi et al. [127], Choi et al. [187]
OsWRKY93	LOC_Os05g40080	*M. oryzae* (P)	Li et al. [188]
OsWRKY104	LOC_Os11g02520	*M. oryzae* (P)	Wang et al. [189]
OsWRKY114	LOC_Os12g02400	*F. fujikuroi* (P), *Xoo* (P)	Son et al. [190], Son et al. [191], Song et al. [192]

OsWRKYs act as positive (P), negative (N), or both positive and negative (PN) regulators of the immune response to the pathogen.

[163]. Although JA and ET increase resistance to *C. suppressalis*, they negatively affect resistance to *N. lugens* [195,196]. Consequently, OsWRKY53-mediated suppression of JA and ET, coupled with H_2O_2 accumulation, leads to increased resistance to *N. lugens* [164]. Interestingly, OsERF3, which promotes the expression of *OsMPK3* and *OsWRKY53*, has an opposite effect to OsWRKY53 in insect resistance [15], suggesting that the interplay between these TFs determines the dominant resistance to diverse pests. OsMPK17 negatively regulates *Xa21*-mediated immunity by upregulating OsWRKY62 [170]. Therefore, the reduction of OsMPK17 abundance in *Xa21* background rice plants in response to *Xoo* infection increases bacterial blight resistance. However, despite the expectation that BWMK1-phosphorylated OsWRKY81, previously referred to as OsWRKY33, plays a role in the MAPK cascade contributing to rice immunity [197], there is no genetic evidence demonstrating that OsWRKY81 regulates a defense response against pathogens.

XA21, a cell-surface receptor-like kinase, serves as an immune receptor recognizing sulfurized RaxX, a microbial molecule exhibiting properties akin to both PAMPs and effectors, thereby triggering defense responses to *Xoo* [198]. Upon perception of the sulfurized RaxX AxYS22, XA21 undergoes proteolytic cleavage, releasing its intracellular kinase domain harboring a nuclear localization sequence [169]. Nuclear translocation of the intracellular kinase domain is crucial for *Xa21*-mediated immunity, and it exclusively interacts with OsWRKY62 in the nucleus [169]. Although OsWRKY42, OsWRKY67, and OsWRKY68 are reported as positive regulators of *Xa21*-mediated immunity [143,178,181], their roles and regulatory mechanisms remain elusive. However, *OsWRKY62* and *OsWRKY76*, suggested to be duplicated genes and functioning as transcriptional repressors, function in both basal defense and *Xa21*-mediated immunity [168,173,174]. Their functions and regulatory mechanisms in rice-pathogen interactions are described in Fig. 6C. OsWRKY62 was identified as the XA1-binding protein that represses immune responses [168]. Alternative splicing is a regulatory mechanism that determines the functions of OsWRKY62 and OsWRKY76. Alternative splicing generates full-length and short-length proteins of OsWRKY62 and OsWRKY76, capable of forming homo- and hetero-oligomers [174]. Full-length OsWRKY62 and OsWRKY76 increase susceptibility to *M. oryzae* and *Xoo* by inducing metabolic and transcriptional alterations [174,175,186]. The *Xoo*-induced full-length OsWRKY62 represses the expression of three *OsGLY II* genes, which are responsible for the degradation of the sugar metabolite methylglyoxal into D-lactate [172]. Thus, the accumulation of methylglyoxal increases susceptibility to *Xoo* by inhibiting the protease activity of OsCDR1. However, the transcriptional repressor activity of full-length OsWRKY62 is compromised in the short alternative form, which interacts with and exerts a dominant-negative effect on full-length OsWRKY62 and OsWRKY76 proteins [174]. The nucleocytoplasmic shuttling and binding proteins are essential for the functional roles of OsWRKY62. OsWRKY62 is localized in the cytosol, but its interaction with OsWRKY76 facilitates its translocation to the nucleus [174]. OsWRKY62, OsWRKY76, and the *M. oryzae* effector AvrPib interact with the nuclear transport receptors OsIMα1a and OsIMα1b, leading their nuclear localization for negative regulation of blast resistance [176]. However, the OsWRKY62-OsWRKY45 heterodimer directly activates the promoter of *DPF* [155]. OsWRKY62 functions as a positive regulator of rice immunity by interacting with OsWRKY45, thereby increasing resistance to *M. oryzae* and *Xoo* through the induction of defense-related genes [155]. Moreover, OsWRKY62 plays a role as a positive or negative regulator of blast resistance, depending on the presence of *Pi9*,

which encodes a typical NLR protein conferring BSDR to *M. oryzae* strains [171]. In non-*Pi9* background rice plant, OsWRKY62 increases innate immunity against *M. oryzae*, but the effector AvrPi9 stabilizes ANIP1, promoting the degradation of OsWRKY62 via the 26S proteasome pathway, thereby enabling the blast fungus to overcome this immune response [171]. However, OsWRKY62 reduces blast resistance in rice plants with a *Pi9* background [171]. Pi9 stabilizes ANIP1 and OsWRKY62 proteins and forms a complex with them to inactivate *Pi9*-mediated immunity, but recognition of AvrPi9 by Pi9 dissociates this complex and activates ETI [171]. These findings suggest that OsWRKYs may function as either negative or positive regulators of rice immunity, depending on the interacting proteins.

The transcriptional activator OsWRKY114 and the transcriptional repressor OsWRKY65 confer BSDR to bacterial blight and bakanae disease. Direct interaction between OsWRKY114 and the W-box and/or the WLE1 motif on the promoters of *OsPR1a* and *CHT4*/*OsPR3* induces their expressions [190]. Overexpression of *OsWRKY114* suppresses both ABA synthesis and ABA signaling, resulting in increased resistance to bacterial blight by directly attenuating ABA signaling and indirectly elevating SA signaling [191]. OsWRKY114 is implicated in the GA and JA signaling pathways via modulating the transcripts of genes associated with these pathways, thereby increasing innate immunity against *F. fujikuroi* [192]. Transcriptomic analysis unveiled that OsWRKY65 modulates a vast array of gene expressions, including those of key regulators of immune responses [177]. Overexpression of *OsWRKY65* leads to the modulation of transcripts related to GA and JA pathways and their endogenous contents, thereby increasing resistance to *F. fujikuroi* [177]. The expressions of susceptibility genes such as *OsSWEET13* and *OsWRKY14* are reduced in *OsWRKY65*-overexpressing plants, while those of *PRs* and SA defense genes are elevated, increasing rice immunity against *Xoo* strains [177]. OsWRKY65 binds to the W-box motif of the *OsSWEE13* promoter, leading to its promoter activity inhibition [177]. Both OsWRKY114 and OsWRKY65 positively regulate disease resistance but impair plant growth and development through their influence on defense-related gene expressions and phytohormone signaling pathways [177,192]. These studies showed that OsWRKYs, acting as transcriptional activators or repressors, may play synthetic roles in the plant growth-defense tradeoff.

The OsWRKY transcriptional regulatory cascades act a central role in modulating rice immunity. The OsWRKY45-OsWRKY13-OsWRKY42 module enhances defense mechanisms against *M. oryzae* [132], while the OsWRKY80-OsWRKY4 module confers resistance to *R. solani* [121]. Recent investigations have elucidated the composition of the OsWRKY transcriptional regulatory network, which consists of OsWRKY6, OsWRKY10, OsWRKY45, OsWRKY47, OsWRKY51, and OsWRKY88, contributing to basal defense and *Xa1*-mediated immunity against *Xoo* [122,127]. XA1 is the first identified typical NLR protein that activates ETI through recognizing the transcriptional activator-like effector of *Xoo* [199]. Through genome-wide expression analyses, Choi et al. identified several OsWRKYs, notably OsWRKY47 and OsWRKY88, potentially implicated in *Xa1*-mediated resistance [187]. Overexpression of either *OsWRKY47* or *OsWRKY88* in rice resulted in elevated transcript levels of *OsPRs* and increased resistance to *M. oryzae* and *Xoo*, respectively [158,187]. They further elucidated the regulatory roles of OsWRKY10 in basal defense and *Xa1*-mediated immunity against *Xoo*, highlighting its capacity to upregulate defense-related genes and directly target *OsWRKY47* [127]. OsWRKY10 activated the promoter of *OsPR1a*, and in collaboration with OsWRKY47, increased the promoter activity of *OsPR1a* [127]. Moreover, OsWRKY51 and OsWRKY88 were identified as upstream regulators activating the OsWRKY10-OsWRKY47 module [127]. OsWRKY6 is also involved in this signaling pathway. OsWRKY6 was shown to directly stimulate the promoters of *OsPR10a* and *OsICS1*, eliciting both basal defense mechanisms and *Xa1*-mediated immunity against *Xoo* [118,122,123]. OsWRKY6 indirectly induced the expression of *OsPR1a* and *OsPR1b* via direct activation of the *OsWRKY45* and *OsWRKY47* promoters [122,148,152]. Im et al. revealed a positive feedback loop wherein OsWRKY45 and OsWRKY47 mutually activate each other's promoters, with OsWRKY45 acting as an auto-activator [122]. Thus, these findings collectively underscore the intricate involvement of various OsWRKY transcription factors in coordinating defense responses against *Xoo*. Recently, Son et al. revealed that 13 *OsWRKYs*, such as *OsWRKY5*, *OsWRKY13*, and *OsWRKY30*, are differentially expressed in *OsWRKY65*-overexpressing plants conferring BSDR [177]. OsWRKY13 induces *OsPR1a* expression and increases resistance to the blast fungus *M. oryzae* and *Xoo* [130]. OsMAPK6-induced *OsWRKY30* increases resistance to *Xoo* and *Xoc* [142]. Given that OsWRKY13 and OsWRKY30 serve as activators of resistance to *Xoo*, the induction of these genes by OsWRKY65 implies their participation in the OsWRKY65 signaling pathway. OsWRKY5 acts as a suppressor of ABA signaling, regulating the expression of multiple genes implicated in both biotic and abiotic stress responses [200]. The 137 genes, such as *OsOPR1*, *OsOPR10*, *OsHI-LOX* and *OsERF101*, are regulated by both OsWRKY5 and OsWRKY65 [177,200]. OPR and LOX2 are enzymes in the JA biosynthesis pathway that function in plant immunity [34,201]. OsERF101 is involved in JA signaling as well as *Xa1*-mediated immunity against *Xoo* [24,202]. These observations suggest that OsWRKY65-mediated suppression of *OsWRKY5* expression increases JA-dependent and effector-triggered immunity. Nevertheless, the complexity of OsWRKY transcriptional regulatory networks remains incompletely understood.

8. Concluding remarks and future perspectives

Plants are continuously challenged by a variety of pathogens, making novel genes that enhance immune responses highly valuable for potential application in crop improvement. Indeed, overexpression or mutagenesis of genes belonged to AP2/ERF, bHLH, bZIP, MYB, NAC, and WRKY superfamily has been shown to enhance disease resistance in rice and other crops. However, TFs involved in various biotic stresses remain largely unidentified in crop plants. Currently, innovative gene-editing technologies are continually being increased based on the groundbreaking CRISPR technology, and thus the construction and screening of CRISPR libraries serve as effective strategies for discovering numerous new genes involved in disease resistance. Moreover, ongoing investigations have elucidated the detailed regulatory mechanisms of these TFs in rice immunity, offering valuable insights into their potential applications in rice and other crops. For instance, a recent study on the diORF translation of *OsWRKY7* provides a valuable approach for increasing specific traits, such as innate immunity, by improving protein stability, which can be applied to other genes in rice and various plants. Therefore, identifying TFs involved in the immune response and unraveling their detailed regulatory mechanisms will provide important methods for plant improvement. However, genes that confer strong defense responses often result in yield penalties. Thus, crop improvement strategies using defense-related genes require additional genetic engineering approaches to selectively activate these genes under specific conditions. Furthermore, current genetic engineering methods face limitations in various crops. Therefore, continued advancements in biotechnology and transformation techniques are essential for the effective application of novel rice resistance genes and their mechanisms to other crop species.

To manage the plant immune response efficiently and flexibly, TFs commonly operate within interconnected networks to respond to the same biotic stress through diverse mechanisms. For instance, in the OsWRKY transcriptional regulatory cascades, TFs form complex networks, such as cascades and feedback loops, to regulate the expression of other TFs, enabling dynamic and coordinated control of innate immunity. Moreover, some TFs influence the activity or function of other TFs at the post-translational level, such as the interaction between OsNAC2 and OsEREBP1, by activating or inactivating them through post-translational modifications (e.g., phosphorylation and ubiquitination) and protein–protein interactions. In addition to these mechanisms, TFs cooperate with each other to efficiently activate resistance to specific biotic stresses by sharing common target genes and engaging in cooperative binding. However, the nature of these relationships remains largely unexplored in rice. Therefore, further studies are essential to elucidate the intricate connections among TFs in rice immunity.

Rice is a vital staple crop in global agriculture; thus, ensuring its sustained cultivation is essential for increasing global food security. Herein, we discuss the diverse roles of six TF superfamilies in rice immunity. By elucidating how these TFs are regulated and their functions as positive and/or negative regulators of pathogen resistance, this review provides candidate genes for rice breeding and offers insights into the mechanisms involved in rice immunity, thereby increasing our understanding and application of TF superfamilies in managing plant diseases.

CRediT authorship contribution statement

Sang Ryeol Park: Writing – review & editing, Writing – original draft. **Yujeong Jeong:** Writing – review & editing, Writing – original draft. **Seungmin Son:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Conceptualization.

Declaration of competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by Research Program for Agricultural Science and Technology Development, Republic of Korea (PJ01570601) and the Fellowship Program (PJ01661001) of the National Institute of Agricultural Sciences, Republic of Korea; Rural Development Administration, Republic of Korea.

Appendix A. Supplementary data

Supplementary data for this article can be found online at https://doi.org/10.1016/j.cj.2024.10.006.

References

[1] T.M. Chaloner, S.J. Gurr, D.P. Bebber, Plant pathogen infection risk tracks global crop yields under climate change, Nat. Clim. Chang. 11 (2021) 710–715.
[2] S. Son, S.R. Park, Climate change impedes plant immunity mechanisms, Front. Plant Sci. 13 (2022) 1032820.
[3] M. Zhang, S. Zhang, Mitogen-activated protein kinase cascades in plant signaling, J. Integr. Plant Biol. 64 (2022) 301–341.
[4] J.M. Zhou, Y. Zhang, Plant immunity: danger perception and signaling, Cell 81 (2020) 978–989.
[5] W. Liu, G.L. Wang, Plant innate immunity in rice: a defense against pathogen infection, Natl. Sci. Rev. 3 (2016) 295–308.
[6] A.E. Douglas, Strategies for enhanced crop resistance to insect pests, Annu. Rev. Plant Biol. 69 (2018) 637–660.
[7] W. Chen, Z. Chen, F. Luo, M. Liao, S. Wei, Z. Yang, J. Yang, RicetissueTFDB: a genome-wide identification of tissue-specific transcription factors in rice, Plant Genome 12 (2019) 170081.
[8] P. Cao, K.H. Jung, D. Choi, D. Hwang, J. Zhu, P.C. Ronald, The rice oligonucleotide array database: an atlas of rice gene expression, Rice 5 (2012) 17.
[9] A.K.N. Chandran, S. Moon, Y.H. Yoo, Y.S. Gho, P. Cao, R. Sharma, M.K. Sharma, P.C. Ronald, K.H. Jung, A web-based tool for the prediction of rice transcription factor function, Database 2019 (2019) baz061.
[10] K.T.X. Vo, M.M. Rahman, M.M. Rahman, K.T.T. Trinh, S.T. Kim, J.S. Jeon, Proteomics and metabolomics studies on the biotic stress responses of rice: an update, Rice 14 (2021) 30.
[11] Z. Xie, T.M. Nolan, H. Jiang, Y. Yin, AP2/ERF transcription factor regulatory networks in hormone and abiotic stress responses in Arabidopsis, Front. Plant Sci. 10 (2019) 228.
[12] N. Ma, P. Sun, Z.Y. Li, F.J. Zhang, X.F. Wang, C.X. You, C.L. Zhang, Z. Zhang, Plant disease resistance outputs regulated by AP2/ERF transcription factor family, Stress Biol. 4 (2024) 2.
[13] H.M. Guo, H.C. Li, S.R. Zhou, H.W. Xue, X.X. Miao, Cis-12-oxo-phytodienoic acid stimulates rice defense response to a piercing-sucking insect, Mol. Plant 7 (2014) 1683–1692.
[14] Y. Ling, Z. Weilin, Genetic and biochemical mechanisms of rice resistance to planthopper, Plant Cell Rep. 35 (2016) 1559–1572.
[15] J. Lu, H. Ju, G. Zhou, C. Zhu, M. Erb, X. Wang, P. Wang, Y. Lou, An EAR-motif-containing ERF transcription factor affects herbivore-induced signaling, defense and resistance in rice, Plant J. 68 (2011) 583–596.
[16] Y. Zhan, X. Sun, G. Rong, C. Hou, Y. Huang, D. Jiang, X. Weng, Identification of two transcription factors activating the expression of OsXIP in rice defence response, BMC Biotechnol. 17 (2017) 26.
[17] Y.H. Cheong, B.C. Moon, J.K. Kim, C.Y. Kim, M.C. Kim, I.H. Kim, C.Y. Park, J.C. Kim, B.O. Park, S.C. Koo, H.W. Yoon, W.S. Chung, C.O. Lim, S.Y. Lee, M.J. Cho, BWMK1, a rice mitogen-activated protein kinase, locates in the nucleus and mediates pathogenesis-related gene expression by activation of a transcription factor, Plant Physiol. 132 (2003) 1961–1972.
[18] X. Xiao, Z. Tang, X. Li, Y. Hong, B. Li, W. Xiao, Z. Gao, D. Lin, C. Li, L. Luo, X. Niu, C. He, Y. Chen, Overexpressing OsMAPK12-1 inhibits plant growth and enhances resistance to bacterial disease in rice, Funct. Plant Biol. 44 (2017) 694–704.
[19] V. Jisha, L. Dampanaboina, J. Vadassery, A. Mithofer, S. Kappara, R. Ramanan, Overexpression of an AP2/ERF type transcription factor OsEREBP1 confers biotic and abiotic stress tolerance in rice, PLoS ONE 10 (2015) e0127831.
[20] Q. Zhong, J. Yu, Y. Wu, X. Yao, C. Mao, X. Meng, F. Ming, Rice transcription factor OsNAC2 maintains the homeostasis of immune responses to bacterial blight, Plant Physiol. 195 (2024) 785–798.
[21] D. Tezuka, A. Kawamata, H. Kato, W. Saburi, H. Mori, R. Imai, The rice ethylene response factor OsERF83 positively regulates disease resistance to *Magnaporthe oryzae*, Plant Physiol. Biochem. 135 (2019) 263–271.
[22] M.J. Wamaitha, R. Yamamoto, H.L. Wong, T. Kawasaki, Y. Kawano, K. Shimamoto, OsRap2.6 transcription factor contributes to rice innate immunity through its interaction with Receptor for Activated Kinase-C 1 (RACK1), Rice 5 (2012) 35.
[23] H. Wang, Y. Bi, Y. Gao, Y. Yan, X. Yuan, X. Xiong, J. Wang, J. Liang, D. Li, F. Song, A pathogen-inducible rice NAC transcription factor ONAC096 contributes to immunity against *Magnaprothe oryzae* and *Xanthomonas oryzae* pv. *oryzae* by direct binding to the promoters of OsRap2.6, OsWRKY62, and OsPAL1, Front. Plant Sci. 12 (2021) 802758.
[24] A. Yoshihisa, S. Yoshimura, M. Shimizu, S. Sato, S. Matsuno, A. Mine, K. Yamaguchi, T. Kawasaki, The rice OsERF101 transcription factor regulates the NLR Xa1-mediated immunity induced by perception of TAL effectors, New Phytol. 236 (2022) 1441–1454.
[25] D. Liu, X. Chen, J. Liu, J. Ye, Z. Guo, The rice ERF transcription factor OsERF922 negatively regulates resistance to *Magnaporthe oryzae* and salt tolerance, J. Exp. Bot. 63 (2012) 3899–3911.
[26] F. Wang, C. Wang, P. Liu, C. Lei, W. Hao, Y. Gao, Y.G. Liu, K. Zhao, Enhanced rice blast resistance by CRISPR/Cas9-targeted mutagenesis of the ERF transcription factor gene OsERF922, PLoS ONE 11 (2016) e0154027.
[27] Y. Zhou, S. Xu, N. Jiang, X. Zhao, Z. Bai, J. Liu, W. Yao, Q. Tang, G. Xiao, C. Lv, K. Wang, X. Hu, J. Tan, Y. Yang, Engineering of rice varieties with enhanced resistances to both blast and bacterial blight diseases via CRISPR/Cas9, Plant Biotechnol. J. 20 (2022) 876–885.
[28] T.T. Tran, A.L. Perez-Quintero, I. Wonni, S.C.D. Carpenter, Y. Yu, L. Wang, J.E. Leach, V. Verdier, S. Cunnac, A.J. Bogdanove, R. Koebnik, M. Hutin, B. Szurek, Functional analysis of African *Xanthomonas oryzae* pv. *oryzae* TALomes reveals a new susceptibility gene in bacterial leaf blight of rice, PLoS Pathog. 14 (2018) e1007092.
[29] X. Sun, Y. Wang, N. Sui, Transcriptional regulation of bHLH during plant response to stress, Biochem. Biophys. Res. Commun. 503 (2018) 397–401.
[30] X. Li, X. Duan, H. Jiang, Y. Sun, Y. Tang, Z. Yuan, J. Guo, W. Liang, L. Chen, J. Yin, H. Ma, J. Wang, D. Zhang, Genome-wide analysis of basic/helix-loop-helix transcription factor family in rice and Arabidopsis, Plant Physiol. 141 (2006) 1167–1184.
[31] Y. Uji, S. Taniguchi, D. Tamaoki, H. Shishido, K. Akimitsu, K. Gomi, Overexpression of OsMYC2 results in the up-regulation of early JA-responsive genes and bacterial blight resistance in rice, Plant Cell Physiol. 57 (2016) 1814–1827.

[32] S. Ogawa, R. Kawahara-Miki, K. Miyamoto, H. Yamane, H. Nojiri, Y. Tsujii, K. Okada, OsMYC2 mediates numerous defence-related transcriptional changes via jasmonic acid signalling in rice, Biochem. Biophys. Res. Commun. 486 (2017) 796–803.
[33] Y. Kong, G. Wang, X. Chen, L. Li, X. Zhang, S. Chen, Y. He, G. Hong, OsPHR2 modulates phosphate starvation-induced OsMYC2 signalling and resistance to *Xanthomonas oryzae* pv. *oryzae*, Plant Cell Environ. 44 (2021) 3432–3444.
[34] J. Qiu, J. Xie, Y. Chen, Z. Shen, H. Shi, N.I. Naqvi, Q. Qian, Y. Liang, Y. Kou, Warm temperature compromises JA-regulated basal resistance to enhance *Magnaporthe oryzae* infection in rice, Mol. Plant 15 (2022) 723–739.
[35] J. Hu, J. Huang, H. Xu, Y. Wang, C. Li, P. Wen, X. You, X. Zhang, G. Pan, Q. Li, H. Zhang, J. He, H. Wu, L. Jiang, H. Wang, Y. Liu, J. Wan, Rice stripe virus suppresses jasmonic acid-mediated resistance by hijacking brassinosteroid signaling pathway in rice, PLoS Pathog. 16 (2020) e1008801.
[36] L. Li, H. Zhang, C. Chen, H. Huang, X. Tan, Z. Wei, J. Li, F. Yan, C. Zhang, J. Chen, Z. Sun, A class of independently evolved transcriptional repressors in plant RNA viruses facilitates viral infection and vector feeding, Proc. Natl. Acad. Sci. U. S. A. 118 (2021).
[37] X. Tan, H. Zhang, Z. Yang, Z. Wei, Y. Li, J. Chen, Z. Sun, NF-YA transcription factors suppress jasmonic acid-mediated antiviral defense and facilitate viral infection in rice, PLoS Pathog. 18 (2022) e1010548.
[38] H. Zhang, F. Wang, W. Song, Z. Yang, L. Li, Q. Ma, X. Tan, Z. Wei, Y. Li, J. Li, F. Yan, J. Chen, Z. Sun, Different viral effectors suppress hormone-mediated antiviral immunity of rice coordinated by OsNPR1, Nat. Commun. 14 (2023) 3011.
[39] Y.S. Dai, D. Liu, W. Guo, Z.X. Liu, X. Zhang, L.L. Shi, D.M. Zhou, L.N. Wang, K. Kang, F.Z. Wang, S.S. Zhao, Y.F. Tan, T. Hu, W. Chen, P. Li, Q.M. Zhou, L.Y. Yuan, Z. Zhang, Y.Q. Chen, W.Q. Zhang, J. Li, L.J. Yu, S. Xiao, Poaceae-specific beta-1,3;1,4-d-glucans link jasmonate signalling to OsLecRK1-mediated defence response during rice-brown planthopper interactions, Plant Biotechnol. J. 21 (2023) 1286–1300.
[40] G. Jin, J. Qi, H. Zu, S. Liu, J. Gershenzon, Y. Lou, I.T. Baldwin, R. Li, Jasmonate-mediated gibberellin catabolism constrains growth during herbivore attack in rice, Plant Cell 35 (2023) 3828–3844.
[41] H.Y. Shin, M.K. You, J.U. Jeung, J.S. Shin, OsMPK3 is a TEY-type rice MAPK in Group C and phosphorylates OsbHLH65, a transcription factor binding to the E-box element, Plant Cell Rep. 33 (2014) 1343–1353.
[42] M. Wang, D. Yang, F. Ma, M. Zhu, Z. Shi, X. Miao, OsHLH61-OsbHLH96 influences rice defense to brown planthopper through regulating the pathogen-related genes, Rice 12 (2019) 9.
[43] C. Yamamura, E. Mizutani, K. Okada, H. Nakagawa, S. Fukushima, A. Tanaka, S. Maeda, T. Kamakura, H. Yamane, H. Takatsuji, M. Mori, Diterpenoid phytoalexin factor, a bHLH transcription factor, plays a central role in the biosynthesis of diterpenoid phytoalexins in rice, Plant J. 84 (2015) 1100–1113.
[44] D. Lieberherr, N.P. Thao, A. Nakashima, K. Umemura, T. Kawasaki, K. Shimamoto, A sphingolipid elicitor-inducible mitogen-activated protein kinase is regulated by the small GTPase OsRac1 and heterotrimeric G-protein in rice, Plant Physiol. 138 (2005) 1644–1652.
[45] L. Chen, S. Hamada, M. Fujiwara, T. Zhu, N.P. Thao, H.L. Wong, P. Krishna, T. Ueda, H. Kaku, N. Shibuya, T. Kawasaki, K. Shimamoto, The Hop/Sti1-Hsp90 chaperone complex facilitates the maturation and transport of a PAMP receptor in rice innate immunity, Cell Host Microbe 7 (2010) 185–196.
[46] S.H. Kim, T. Oikawa, J. Kyozuka, H.L. Wong, K. Umemura, M. Kishi-Kaboshi, A. Takahashi, Y. Kawano, T. Kawasaki, K. Shimamoto, The bHLH Rac Immunity1 (RAI1) is activated by OsRac1 via OsMAPK3 and OsMAPK6 in rice immunity, Plant Cell Physiol. 53 (2012) 740–754.
[47] F. Meng, C. Yang, J. Cao, H. Chen, J. Pang, Q. Zhao, Z. Wang, Z. Qing Fu, J. Liu, A bHLH transcription activator regulates defense signaling by nucleo-cytosolic trafficking in rice, J. Integr. Plant Biol. 62 (2020) 1552–1573.
[48] J. Liu, Y. Shen, H. Cao, K. He, Z. Chu, N. Li, OsbHLH057 targets the AATCA cis-element to regulate disease resistance and drought tolerance in rice, Plant Cell Rep. 41 (2022) 1285–1299.
[49] T. Onohata, K. Gomi, Overexpression of jasmonate-responsive OsbHLH034 in rice results in the induction of bacterial blight resistance via an increase in lignin biosynthesis, Plant Cell Rep. 39 (2020) 1175–1184.
[50] J.H. Im, S. Son, W.C. Kim, K. Kim, N. Mitsuda, J.H. Ko, K.H. Han, Jasmonate activates secondary cell wall biosynthesis through MYC2-MYB46 module, Plant J. 117 (2024) 1099–1114.
[51] Q. Zhang, R. Teng, Z. Yuan, S. Sheng, Y. Xiao, H. Deng, W. Tang, F. Wang, Integrative transcriptomic analysis deciphering the role of rice bHLH transcription factor Os04g0301500 in mediating responses to biotic and abiotic stresses, Front. Plant Sci. 14 (2023) 1266242.
[52] M. Jakoby, B. Weisshaar, W. Droge-Laser, J. Vicente-Carbajosa, J. Tiedemann, T. Kroj, F. Parcy, Z.I.P.R.G. b, bZIP transcription factors in Arabidopsis, Trends Plant Sci. 7 (2002) 106-111.
[53] M.S. Alves, S.P. Dadalto, A.B. Goncalves, G.B. De Souza, V.A. Barros, L.G. Fietto, Plant bZIP transcription factors responsive to pathogens: a review, Int. J. Mol. Sci. 14 (2013) 7815–7828.
[54] A. Hussain, R. Tariq, A. Aslam, H. Saleem, H. Mumtaz, I. Khan, M. Sajid, H. Basim, Unboxing the bZIP transcription factor family exhibiting their role under cold and salt stresses in indica rice, Plant Stress 10 (2023) 100299.
[55] A. Nijhawan, M. Jain, A.K. Tyagi, J.P. Khurana, Genomic survey and gene expression analysis of the basic leucine zipper transcription factor family in rice, Plant Physiol. 146 (2008) 333–350.
[56] C. Lu, X. Liu, Y. Tang, Y. Fu, J. Zhang, L. Yang, P. Li, Z. Zhu, P. Dong, A comprehensive review of TGA transcription factors in plant growth, stress responses, and beyond, Int. J. Biol. Macromol. 258 (2024) 128880.
[57] M. Chern, H.A. Fitzgerald, P.E. Canlas, D.A. Navarre, P.C. Ronald, Overexpression of a rice NPR1 homolog leads to constitutive activation of defense response and hypersensitivity to light, Mol. Plant-Microbe Interact. 18 (2005) 511–520.
[58] Y. Yuan, S. Zhong, Q. Li, Z. Zhu, Y. Lou, L. Wang, J. Wang, M. Wang, Q. Li, D. Yang, Z. He, Functional analysis of rice NPR1-like genes reveals that OsNPR1/NH1 is the rice orthologue conferring disease resistance with enhanced herbivore susceptibility, Plant Biotechnol. J. 5 (2007) 313–324.
[59] W. Bai, M. Chern, D. Ruan, P.E. Canlas, W.H. Sze-To, P.C. Ronald, Enhanced disease resistance and hypersensitivity to BTH by introduction of an NH1/OsNPR1 paralog, Plant Biotechnol. J. 9 (2011) 205–215.
[60] M. Chern, H.A. Fitzgerald, R.C. Yadav, P.E. Canlas, X. Dong, P.C. Ronald, Evidence for a disease-resistance pathway in rice similar to the NPR1-mediated signaling pathway in Arabidopsis, Plant J. 27 (2001) 101–113.
[61] M. Chern, W. Bai, D. Ruan, T. Oh, X. Chen, P.C. Ronald, Interaction specificity and coexpression of rice NPR1 homologs 1 and 3 (NH1 and NH3), TGA transcription factors and Negative Regulator of Resistance (NRR) proteins, BMC Genomics 15 (2014) 461.
[62] H.A. Fitzgerald, P.E. Canlas, M.S. Chern, P.C. Ronald, Alteration of TGA factor activity in rice results in enhanced tolerance to *Xanthomonas oryzae* pv. *oryzae*, Plant J. 43 (2005) 335–347.
[63] Y. Niu, X. Huang, Z. He, Q. Zhang, H. Meng, H. Shi, B. Feng, Y. Zhou, J. Zhang, G. Lu, Z. Wang, W. Zhang, D. Tang, M. Wang, Phosphorylation of OsTGA5 by casein kinase II compromises its suppression of defense-related gene transcription in rice, Plant Cell 34 (2022) 3425–3442.
[64] S.J. Moon, H.J. Park, T.H. Kim, J.W. Kang, J.Y. Lee, J.H. Cho, J.H. Lee, D.S. Park, M. O. Byun, B.G. Kim, D. Shin, OsTGA2 confers disease resistance to rice against leaf blight by regulating expression levels of disease related genes via interaction with NH1, PLoS ONE 13 (2018) e0206910.
[65] Y. Yin, Q. Zhu, S. Dai, C. Lamb, R.N. Beachy, RF2a, a bZIP transcriptional activator of the phloem-specific rice tungro bacilliform virus promoter, functions in vascular development, EMBO J. 16 (1997) 5247–5259.
[66] S. Dai, Z. Zhang, S. Chen, R.N. Beachy, RF2b, a rice bZIP transcription activator, interacts with RF2a and is involved in symptom development of rice tungro disease, Proc. Natl. Acad. Sci. U. S. A. 101 (2004) 687–692.
[67] S. Dai, X. Wei, A.A. Alfonso, L. Pei, U.G. Duque, Z. Zhang, G.M. Babb, R.N. Beachy, Transgenic rice plants that overexpress transcription factors RF2a and RF2b are tolerant to rice tungro virus replication and disease, Proc. Natl. Acad. Sci. U. S. A. 105 (2008) 21012–21016.
[68] J.H. Im, S.G. Lee, E. Lee, S.R. Park, I. Ahn, D.-J. Hwang, OsbZIP75 positively regulates plant defense against the bacterial leaf blight pathogen *Xanthomonas oryzae* pv. *oryzae*, Plant Biotechnol. Rep. 13 (2019) 645–651.
[69] C.H. Park, S. Chen, G. Shirsekar, B. Zhou, C.H. Khang, P. Songkumarn, A.J. Afzal, Y. Ning, R. Wang, M. Bellizzi, B. Valent, G.L. Wang, The Magnaporthe oryzae effector AvrPiz-t targets the RING E3 ubiquitin ligase APIP6 to suppress pathogen-associated molecular pattern-triggered immunity in rice, Plant Cell 24 (2012) 4748–4762.
[70] R. Wang, Y. Ning, X. Shi, F. He, C. Zhang, J. Fan, N. Jiang, Y. Zhang, T. Zhang, Y. Hu, M. Bellizzi, G.L. Wang, Immunity to rice blast disease by suppression of effector-triggered necrosis, Curr. Biol. 26 (2016) 2399–2411.
[71] F. Zhang, H. Fang, M. Wang, F. He, H. Tao, R. Wang, J. Long, J. Wang, G.L. Wang, Y. Ning, APIP5 functions as a transcription factor and an RNA-binding protein to modulate cell death and immunity in rice, Nucleic Acids Res. 50 (2022) 5064–5079.
[72] A. Sugio, B. Yang, T. Zhu, F.F. White, Two type III effector genes of *Xanthomonas oryzae* pv. *oryzae* control the induction of the host genes OsTFIIAgamma1 and OsTFX1 during bacterial blight of rice, Proc. Natl. Acad. Sci. U. S. A. 104 (2007) 10720-10725.
[73] J. Hu, J. Zhou, X. Peng, H. Xu, C. Liu, B. Du, H. Yuan, L. Zhu, G. He, The Bphi008a gene interacts with the ethylene pathway and transcriptionally regulates MAPK genes in the response of rice to brown planthopper feeding, Plant Physiol. 156 (2011) 856–872.
[74] D. Liang, J. Yu, T. Song, R. Zhang, Y. Du, M. Yu, H. Cao, X. Pan, J. Qiao, Y. Liu, Z. Qi, Y. Liu, Genome-wide prediction and analysis of *Oryza* species NRP genes in rice blast resistance, Int. J. Mol. Sci. 23 (2022) 11967.
[75] D. Liu, S. Shi, Z. Hao, W. Xiong, M. Luo, OsbZIP81, A homologue of Arabidopsis VIP1, may positively regulate JA levels by directly targetting the genes in JA signaling and metabolism pathway in rice, Int. J. Mol. Sci. 20 (2019) 2360.
[76] Y. Wu, J. Wen, Y. Xia, L. Zhang, H. Du, Evolution and functional diversification of R2R3-MYB transcription factors in plants, Hortic. Res. 9 (2022) uhac058.
[77] R.L. Mu, Y.R. Cao, Y.F. Liu, G. Lei, H.F. Zou, Y. Liao, H.W. Wang, W.K. Zhang, B. Ma, J.Z. Du, M. Yuan, J.S. Zhang, S.Y. Chen, An R2R3-type transcription factor gene AtMYB59 regulates root growth and cell cycle progression in Arabidopsis, Cell Res. 19 (2009) 1291–1304.
[78] F. Xie, C. Chen, J. Chen, J. Chen, Q. Hua, K. Shah, Z. Zhang, J. Zhao, G. Hu, J. Chen, Y. Qin, Betalain biosynthesis in red pulp pitaya is regulated via HuMYB132: a R-R type MYB transcription factor, BMC Plant Biol. 23 (2023) 28.
[79] P. Zhang, T. Wang, L. Cao, Z. Jiao, L. Ku, D. Dou, Z. Liu, J. Fu, X. Xie, Y. Zhu, L. Chong, L. Wei, Molecular mechanism analysis of ZmRL6 positively regulating drought stress tolerance in maize, Stress Biol. 3 (2023) 47.
[80] L. Kang, Y. Teng, Q. Cen, Y. Fang, Q. Tian, X. Zhang, H. Wang, X. Zhang, D. Xue, Genome-wide identification of R2R3-MYB transcription factor and expression analysis under abiotic stress in rice, Plants (Basel) 11 (2022) 1928.

[81] W. Li, Z. Zhu, M. Chern, J. Yin, C. Yang, L. Ran, M. Cheng, M. He, K. Wang, J. Wang, X. Zhou, X. Zhu, Z. Chen, J. Wang, W. Zhao, B. Ma, P. Qin, W. Chen, Y. Wang, J. Liu, W. Wang, X. Wu, P. Li, J. Wang, L. Zhu, S. Li, X. Chen, A natural allele of a transcription factor in rice confers broad-spectrum blast resistance, Cell 170 (2017) 114–126.
[82] M. Kishi-Kaboshi, S. Seo, A. Takahashi, H. Hirochika, The MAMP-responsive MYB transcription factors MYB30, MYB55 and MYB110 activate the HCAA synthesis pathway and enhance immunity in rice, Plant Cell Physiol. 59 (2018) 903–915.
[83] J. He, Y. Liu, D. Yuan, M. Duan, Y. Liu, Z. Shen, C. Yang, Z. Qiu, D. Liu, P. Wen, J. Huang, D. Fan, S. Xiao, Y. Xin, X. Chen, L. Jiang, H. Wang, L. Yuan, J. Wan, An R2R3 MYB transcription factor confers brown planthopper resistance by regulating the phenylalanine ammonia-lyase pathway in rice, Proc. Natl. Acad. Sci. U. S. A. 117 (2020) 271–277.
[84] W. Li, K. Wang, M. Chern, Y. Liu, Z. Zhu, J. Liu, X. Zhu, J. Yin, L. Ran, J. Xiong, K. He, L. Xu, M. He, J. Wang, J. Liu, Y. Bi, H. Qing, M. Li, K. Hu, L. Song, L. Wang, T. Qi, Q. Hou, W. Chen, Y. Li, W. Wang, X. Chen, Sclerenchyma cell thickening through enhanced lignification induced by OsMYB30 prevents fungal penetration of rice leaves, New Phytol. 226 (2020) 1850–1863.
[85] J.F. Chen, Z.X. Zhao, Y. Li, T.T. Li, Y. Zhu, X.M. Yang, S.X. Zhou, H. Wang, J.Q. Zhao, M. Pu, H. Feng, J. Fan, J.W. Zhang, Y.Y. Huang, W.M. Wang, Fine-tuning roles of Osa-miR159a in rice immunity against *Magnaporthe oryzae* and development, Rice 14 (2021) 26.
[86] Y. Shen, G. Yang, X. Miao, Z. Shi, OsmiR159 modulate BPH resistance through regulating G-protein γ subunit GS3 gene in rice, Rice 16 (2023) 30.
[87] H. Zhang, B. Sun, W. Wu, Y. Li, Z. Yin, C. Lu, H. Zhao, L. Kong, X. Ding, The MYB transcription factor OsMYBxoc1 regulates resistance to *Xoc* through directly repressing transcription of iron transport gene OsNRAMP5 in rice, Plant Commun. (2024) 100859.
[88] S. Pooja, K. Sweta, A. Mohanapriya, C. Sudandiradoss, R. Siva, K.M. Gothandam, S. Babu, Homotypic clustering of OsMYB4 binding site motifs in promoters of the rice genome and cellular-level implications on sheath blight disease resistance, Gene 561 (2015) 209–218.
[89] C. Vannini, M. Iriti, M. Bracale, F. Locatelli, F. Faoro, P. Croce, R. Pirona, A. Di Maro, I. Coraggio, A. Genga, The ectopic expression of the rice Osmyb4 gene in Arabidopsis increases tolerance to abiotic, environmental and biotic stresses, Physiol. Mol. Plant Pathol. 69 (2006) 26–42.
[90] C. Vannini, M. Campa, M. Iriti, A. Genga, F. Faoro, S. Carravieri, G.L. Rotino, M. Rossoni, A. Spinardi, M. Bracale, Evaluation of transgenic tomato plants ectopically expressing the rice Osmyb4 gene, Plant Sci. 173 (2007) 231–239.
[91] W.L. Cao, R.Z. Chu, Y. Zhang, J. Luo, Y.Y. Su, L.J. Xie, H.S. Zhang, J.F. Wang, Y.M. Bao, OsJAMyb, a R2R3-type MYB transcription factor, enhanced blast resistance in transgenic rice, Physiol. Mol. Plant Pathol. 92 (2015) 154–160.
[92] W. Yang, J. Zhao, S. Zhang, L. Chen, T. Yang, J. Dong, H. Fu, Y. Ma, L. Zhou, J. Wang, W. Liu, Q. Liu, B. Liu, Genome-wide association mapping and gene expression analysis reveal the negative role of OsMYB21 in regulating bacterial blight resistance in rice, Rice 14 (2021) 58.
[93] A.N. Olsen, H.A. Ernst, L.L. Leggio, K. Skriver, NAC transcription factors: structurally distinct, functionally diverse, Trends Plant Sci. 10 (2005) 79–87.
[94] S. Puranik, P.P. Sahu, P.S. Srivastava, M. Prasad, NAC proteins: regulation and role in stress tolerance, Trends Plant Sci. 17 (2012) 369–381.
[95] L.S. Tran, K. Nakashima, Y. Sakuma, S.D. Simpson, Y. Fujita, K. Maruyama, M. Fujita, M. Seki, K. Shinozaki, K. Yamaguchi-Shinozaki, Isolation and functional analysis of Arabidopsis stress-inducible NAC transcription factors that bind to a drought-responsive cis-element in the early responsive to dehydration stress 1 promoter, Plant Cell 16 (2004) 2481–2498.
[96] Y. Ogo, T. Kobayashi, R. Nakanishi Itai, H. Nakanishi, Y. Kakei, M. Takahashi, S. Toki, S. Mori, N.K. Nishizawa, A novel NAC transcription factor, IDEF2, that recognizes the iron deficiency-responsive element 2 regulates the genes involved in iron homeostasis in plants, J. Biol. Chem. 283 (2008) 13407–13417.
[97] A. Wu, A.D. Allu, P. Garapati, H. Siddiqui, H. Dortay, M.I. Zanor, M.A. Asensi-Fabado, S. Munne-Bosch, C. Antonio, T. Tohge, A.R. Fernie, K. Kaufmann, G.P. Xue, B. Mueller-Roeber, S. Balazadeh, JUNGBRUNNEN1, a reactive oxygen species-responsive NAC transcription factor, regulates longevity in Arabidopsis, Plant Cell 24 (2012) 482–506.
[98] Y. Ye, K. Wu, J. Chen, Q. Liu, Y. Wu, B. Liu, X. Fu, OsSND2, a NAC family transcription factor, is involved in secondary cell wall biosynthesis through regulating MYBs expression in rice, Rice 11 (2018) 36.
[99] X. Yuan, H. Wang, J. Cai, Y. Bi, D. Li, F. Song, Rice NAC transcription factor ONAC066 functions as a positive regulator of drought and oxidative stress response, BMC Plant Biol. 19 (2019) 278.
[100] Z. Zhang, J. Dong, C. Ji, Y. Wu, J. Messing, NAC-type transcription factors regulate accumulation of starch and protein in maize seeds, Proc. Natl. Acad. Sci. U. S. A. 116 (2019) 11223–11228.
[101] J. Wang, Z. Chen, Q. Zhang, S. Meng, C. Wei, The NAC transcription factors OsNAC20 and OsNAC26 regulate starch and storage protein synthesis, Plant Physiol. 184 (2020) 1775–1791.
[102] Y. Sakuraba, D. Kim, S.H. Han, S.H. Kim, W. Piao, S. Yanagisawa, G. An, N.C. Paek, Multilayered regulation of membrane-bound ONAC054 is essential for abscisic acid-induced leaf senescence in rice, Plant Cell 32 (2020) 630–649.
[103] Y. Bi, H. Wang, X. Yuan, Y. Yan, D. Li, F. Song, The NAC transcription factor ONAC083 negatively regulates rice immunity against *Magnaporthe oryzae* by directly activating transcription of the RING-H2 gene OsRFPH2-6, J. Integr. Plant Biol. 65 (2023) 854–875.
[104] K. Nakashima, L.S. Tran, D. Van Nguyen, M. Fujita, K. Maruyama, D. Todaka, Y. Ito, N. Hayashi, K. Shinozaki, K. Yamaguchi-Shinozaki, Functional analysis of a NAC-type transcription factor OsNAC6 involved in abiotic and biotic stress-responsive gene expression in rice, Plant J. 51 (2007) 617–630.
[105] T. Kaneda, Y. Taga, R. Takai, M. Iwano, H. Matsui, S. Takayama, A. Isogai, F.S. Che, The transcription factor OsNAC4 is a key positive regulator of plant hypersensitive cell death, EMBO J. 28 (2009) 926–936.
[106] L. Sun, H. Zhang, D. Li, L. Huang, Y. Hong, X.S. Ding, R.S. Nelson, X. Zhou, F. Song, Functions of rice NAC transcriptional factors, ONAC122 and ONAC131, in defense responses against *Magnaporthe grisea*, Plant Mol, Biol. 81 (2013) 41–56.
[107] N. Yokotani, T. Tsuchida-Mayama, H. Ichikawa, N. Mitsuda, M. Ohme-Takagi, H. Kaku, E. Minami, Y. Nishizawa, OsNAC111, a blast disease-responsive transcription factor in rice, positively regulates the expression of defense-related genes, Mol. Plant-Microbe. Interact. 27 (2014) 1027–1034.
[108] S.R. Park, H.S. Kim, K.S. Lee, D.J. Hwang, S.C. Bae, I.P. Ahn, S.H. Lee, S.T. Kim, Overexpression of rice NAC transcription factor OsNAC58 on increased resistance to bacterial leaf blight, J. Plant Biotechnol. 44 (2017) 149–155.
[109] Z. Wang, Y. Xia, S. Lin, Y. Wang, B. Guo, X. Song, S. Ding, L. Zheng, R. Feng, S. Chen, Y. Bao, C. Sheng, X. Zhang, J. Wu, D. Niu, H. Jin, H. Zhao, Osa-miR164a targets OsNAC60 and negatively regulates rice immunity against the blast fungus *Magnaporthe oryzae*, Plant J. 95 (2018) 584–597.
[110] S. Son, G. Song, S. Nam, G. Lee, J. Im, K.S. Lee, Y.J. Park, E.J. Suh, S.R. Park, CRISPR/Cas9-mediated mutagenesis of rice NAC transcription factor genes results in altered innate immunity, Plant Physiol. 195 (2024) 1138–1142.
[111] Q. Liu, S. Yan, W. Huang, J. Yang, J. Dong, S. Zhang, J. Zhao, T. Yang, X. Mao, X. Zhu, B. Liu, NAC transcription factor ONAC066 positively regulates disease resistance by suppressing the ABA signaling pathway in rice, Plant Mol. Biol. 98 (2018) 289–302.
[112] X. Yuan, H. Wang, Y. Bi, Y. Yan, Y. Gao, X. Xiong, J. Wang, D. Li, F. Song, ONAC066, A stress-responsive NAC transcription activator, positively contributes to rice immunity against *Magnaprothe oryzae* through modulating expression of OsWRKY62 and three cytochrome P450 genes, Front. Plant. Sci. 12 (2021) 749186.
[113] M. Yoshii, T. Shimizu, M. Yamazaki, T. Higashi, A. Miyao, H. Hirochika, T. Omura, Disruption of a novel gene for a NAC-domain protein in rice confers resistance to rice dwarf virus, Plant J. 57 (2009) 615–625.
[114] M. Yoshii, M. Yamazaki, R. Rakwal, M. Kishi-Kaboshi, A. Miyao, H. Hirochika, The NAC transcription factor RIM1 of rice is a new regulator of jasmonate signaling, Plant J. 61 (2010) 804–815.
[115] K. Satoh, T. Shimizu, H. Kondoh, A. Hiraguri, T. Sasaya, I.R. Choi, T. Omura, S. Kikuchi, Relationship between symptoms and gene expression induced by the infection of three strains of Rice dwarf virus, PLoS One 6 (2011) e18094.
[116] T. Eulgem, P.J. Rushton, S. Robatzek, I.E. Somssich, The WRKY superfamily of plant transcription factors, Trends Plant Sci. 5 (2000) 199–206.
[117] I. Ciolkowski, D. Wanke, R.P. Birkenbihl, I.E. Somssich, Studies on DNA-binding selectivity of WRKY transcription factors lend structural clues into WRKY-domain function, Plant Mol. Biol. 68 (2008) 81–92.
[118] C. Choi, S.H. Hwang, I.R. Fang, S.I. Kwon, S.R. Park, I. Ahn, J.B. Kim, D.J. Hwang, Molecular characterization of *Oryza sativa* WRKY6, which binds to W-box-like element 1 of the *Oryza sativa* pathogenesis-related (PR) 10a promoter and confers reduced susceptibility to pathogens, New Phytol. 208 (2015) 846–859.
[119] J.Y. Jang, C.H. Choi, D.J. Hwang, The WRKY superfamily of rice transcription factors, Plant Pathol. J. 26 (2010) 110–114.
[120] H. Wang, J. Meng, X. Peng, X. Tang, P. Zhou, J. Xiang, X. Deng, Rice WRKY4 acts as a transcriptional activator mediating defense responses toward *Rhizoctonia solani*, the causing agent of rice sheath blight, Plant Mol. Biol. 89 (2015) 157–171.
[121] X. Peng, H. Wang, J.C. Jang, T. Xiao, H. He, D. Jiang, X. Tang, OsWRKY80-OsWRKY4 module as a positive regulatory circuit in rice resistance against *Rhizoctonia solani*, Rice 9 (2016) 63.
[122] J.H. Im, C. Choi, S.R. Park, D.J. Hwang, The OsWRKY6 transcriptional cascade functions in basal defense and Xa1-mediated defense of rice against *Xanthomonas oryzae* pv. *oryzae*, Planta 255 (2022) 47.
[123] J.H. Im, C. Choi, M.Y. Jung, S.R. Park, D.J. Hwang, The OsICS1 is directly regulated by OsWRKY6 and increases resistance against *Xanthomonas oryzae* pv. *oryzae*, Planta 259 (2024) 124.
[124] W. Tun, J. Yoon, K.T.X. Vo, L.H. Cho, T.V. Hoang, X. Peng, E.J. Kim, K. Win, S.W. Lee, K.H. Jung, J.S. Jeon, G. An, Sucrose preferentially promotes expression of OsWRKY7 and OsPR10a to enhance defense response to blast fungus in rice, Front. Plant Sci. 14 (2023) 1117023.
[125] S. Son, G. Song, J.H. Im, OsWRKY7 contributes to pattern-triggered immunity against *Xanthomonas oryzae* pv. *oryza*e, Biochem. Biophys. Res. Commun. 700 (2024) 149568.
[126] C. Zheng, J. Zhou, X. Yuan, E. Zheng, X. Liu, W. Cui, C. Yan, Y. Wu, W. Ruan, K. Yi, J. Chen, X. Wang, Elevating plant immunity by translational regulation of a rice WRKY transcription factor, Plant Biotechnol. J. 22 (2024) 1033–1048.
[127] N. Choi, J.H. Im, E. Lee, J. Lee, C. Choi, S.R. Park, D.J. Hwang, WRKY10 transcriptional regulatory cascades in rice are involved in basal defense and Xa1-mediated resistance, J. Exp. Bot. 71 (2020) 3735–3748.
[128] L. Wang, J. Fu, Q. Shen, Q. Wang, OsWRKY10 extensively activates multiple rice diterpenoid phytoalexin biosynthesis genes to enhance rice blast resistance, Plant J. 115 (2023) 758–771.

[129] H. Lee, J. Cha, C. Choi, N. Choi, H.S. Ji, S.R. Park, S. Lee, D.J. Hwang, Rice WRKY11 plays a role in pathogen defense and drought tolerance, Rice 11 (2018) 1–12.
[130] D. Qiu, J. Xiao, X. Ding, M. Xiong, M. Cai, Y. Cao, X. Li, C. Xu, S. Wang, OsWRKY13 mediates rice disease resistance by regulating defense-related genes in salicylate- and jasmonate-dependent signaling, Mol. Plant-Microbe Interact. 20 (2007) 492–499.
[131] J. Xiao, H. Cheng, X. Li, J. Xiao, C. Xu, S. Wang, Rice WRKY13 regulates cross talk between abiotic and biotic stress signaling pathways by selective binding to different cis-elements, Plant Physiol. 163 (2013) 1868–1882.
[132] H. Cheng, H. Liu, Y. Deng, J. Xiao, X. Li, S. Wang, The WRKY45-2 WRKY13 WRKY42 transcriptional regulatory cascade is required for rice resistance to fungal pathogen, Plant Physiol. 167 (2015) 1087–1099.
[133] Q. Qin, G. Li, L. Jin, Y. Huang, Y. Wang, C. Wei, Z. Xu, Z. Yang, H. Wang, Y. Li, Auxin response factors (ARFs) differentially regulate rice antiviral immune response against rice dwarf virus, PLoS Pathog. 16 (2020) e1009118.
[134] D. Du, C. Zhang, Y. Xing, X. Lu, L. Cai, H. Yun, Q. Zhang, Y. Zhang, X. Chen, M. Liu, X. Sang, Y. Ling, Z. Yang, Y. Li, B. Lefebvre, G. He, The CC-NB-LRR OsRLR1 mediates rice disease resistance through interaction with OsWRKY19, Plant Biotechnol. J. 19 (2021) 1052–1064.
[135] P. Abbruscato, T. Nepusz, L. Mizzi, M. Del Corvo, P. Morandini, I. Fumasoni, C. Michel, A. Paccanaro, E. Guiderdoni, U. Schaffrath, J.B. Morel, P. Piffanelli, O. Faivre-Rampant, OsWRKY22, a monocot WRKY gene, plays a role in the resistance response to blast, Mol. Plant Pathol. 13 (2012) 828–841.
[136] N. Yokotani, M. Shikata, H. Ichikawa, N. Mitsuda, M. Ohme-Takagi, E. Minami, Y. Nishizawa, OsWRKY24, a blast-disease responsive transcription factor, positively regulates rice disease resistance, J. Gen. Plant Pathol. 84 (2018) 85–91.
[137] T. Chujo, K. Miyamoto, T. Shimogawa, T. Shimizu, Y. Otake, N. Yokotani, Y. Nishizawa, N. Shibuya, H. Nojiri, H. Yamane, E. Minami, K. Okada, OsWRKY28, a PAMP-responsive transrepressor, negatively regulates innate immune responses in rice against rice blast fungus, Plant Mol. Biol. 82 (2013) 23–37.
[138] X. Peng, Y. Hu, X. Tang, P. Zhou, X. Deng, H. Wang, Z. Guo, Constitutive expression of rice WRKY30 gene increases the endogenous jasmonic acid accumulation, PR gene expression and resistance to fungal pathogens in rice, Planta 236 (2012) 1485–1498.
[139] H. Lee, Y.J. Ko, J.Y. Cha, S.R. Park, I. Ahn, D.J. Hwang, The C-terminal region of Os WRKY30 is sufficient to confer enhanced resistance to pathogen and activate the expression of defense-related genes, Plant Biotechnol. Rep. 7 (2013) 221–230.
[140] M. Han, H.S. Ryu, C.Y. Kim, D.S. Park, Y.K. Ahn, J.S. Jeon, OsWRKY30 is a transcription activator that enhances rice resistance to the *Xanthomonas oryzae* pathovar *oryzae*, J. Plant Biol. 56 (2013) 258–265.
[141] S.K. Jalmi, A.K. Sinha, Functional involvement of a mitogen activated protein kinase module, OsMKK3-OsMPK7-OsWRK30 in mediating resistance against *Xanthomonas oryzae* in rice, Sci. Rep. 6 (2016) 37974.
[142] L. Wang, J. Chen, Y. Zhao, S. Wang, M. Yuan, OsMAPK6 phosphorylates a zinc finger protein OsLIC to promote downstream OsWRKY30 for rice resistance to bacterial blight and leaf streak, J. Integr. Plant Biol. 64 (2022) 1116–1130.
[143] T. Wang, Z. Zhu, Y. Chen, Y. Liu, G. Yan, S. Xu, T. Zhang, J. Ma, S. Dou, L. Li, Rice OsWRKY42 is a novel element in Xa21-mediated resistance pathway against bacterial leaf blight, Chin. Bull. Bot. 56 (2021) 687.
[144] M. Shimono, S. Sugano, A. Nakayama, C.J. Jiang, K. Ono, S. Toki, H. Takatsuji, Rice WRKY45 plays a crucial role in benzothiadiazole-inducible blast resistance, Plant Cell 19 (2007) 2064–2076.
[145] Z. Tao, H. Liu, D. Qiu, Y. Zhou, X. Li, C. Xu, S. Wang, A pair of allelic WRKY genes play opposite roles in rice-bacteria interactions, Plant Physiol. 151 (2009) 936–948.
[146] M. Shimono, H. Koga, A. Akagi, N. Hayashi, S. Goto, M. Sawada, T. Kurihara, A. Matsushita, S. Sugano, C.J. JIANG, Rice WRKY45 plays important roles in fungal and bacterial disease resistance, Mol. Plant Pathol. 13 (2012) 83–94.
[147] A. Matsushita, H. Inoue, S. Goto, A. Nakayama, S. Sugano, N. Hayashi, H. Takatsuji, Nuclear ubiquitin proteasome degradation affects WRKY45 function in the rice defense program, Plant J. 73 (2013) 302–313.
[148] C.J. Jiang, M. Shimono, S. Sugano, M. Kojima, X. Liu, H. Inoue, H. Sakakibara, H. Takatsuji, Cytokinins act synergistically with salicylic acid to activate defense gene expression in rice, Mol. Plant-Microbe Interact. 26 (2013) 287–296.
[149] Y. Ueno, R. Yoshida, M. Kishi-Kaboshi, A. Matsushita, C.J. Jiang, S. Goto, A. Takahashi, H. Hirochika, H. Takatsuji, MAP kinases phosphorylate rice WRKY45, Plant Signal. Behav. 8 (2013) e24510.
[150] H. Inoue, N. Hayashi, A. Matsushita, L. Xinqiong, A. Nakayama, S. Sugano, C.J. Jiang, H. Takatsuji, Blast resistance of CC-NB-LRR protein Pb1 is mediated by WRKY45 through protein-protein interaction, Proc. Natl. Acad. Sci. U. S. A. 110 (2013) 9577–9582.
[151] A. Nakayama, S. Fukushima, S. Goto, A. Matsushita, M. Shimono, S. Sugano, C. J. Jiang, A. Akagi, M. Yamazaki, H. Inoue, H. Takatsuji, Genome-wide identification of WRKY45-regulated genes that mediate benzothiadiazole-induced defense responses in rice, BMC Plant Biol. 13 (2013) 150.
[152] S. Goto, F. Sasakura-Shimoda, M. Suetsugu, M.G. Selvaraj, N. Hayashi, M. Yamazaki, M. Ishitani, M. Shimono, S. Sugano, A. Matsushita, T. Tanabata, H. Takatsuji, Development of disease-resistant rice by optimized expression of WRKY45, Plant Biotechnol. J. 13 (2015) 753–765.
[153] Y. Ueno, R. Yoshida, M. Kishi-Kaboshi, A. Matsushita, C.J. Jiang, S. Goto, A. Takahashi, H. Hirochika, H. Takatsuji, Abiotic stresses antagonize the rice defence pathway through the tyrosine-dephosphorylation of OsMPK6, PLoS Pathog. 11 (2015) e1005231.
[154] J. Huangfu, J. Li, R. Li, M. Ye, P. Kuai, T. Zhang, Y. Lou, The transcription factor OsWRKY45 negatively modulates the resistance of rice to the brown planthopper *Nilaparvata lugens*, Int. J. Mol. Sci. 17 (2016) 697.
[155] S. Fukushima, M. Mori, S. Sugano, H. Takatsuji, Transcription factor WRKY62 plays a role in pathogen defense and hypoxia-responsive gene expression in rice, Plant Cell Physiol. 57 (2016) 2541–2551.
[156] Y. Uji, K. Kashihara, H. Kiyama, S. Mochizuki, K. Akimitsu, K. Gomi, Jasmonic acid-induced VQ-motif-containing protein OsVQ13 influences the OsWRKY45 signaling pathway and grain size by associating with OsMPK6 in rice, Int. J. Mol. Sci. 20 (2019) 2917.
[157] Y. Wu, Y. Fu, Z. Zhu, Q. Hu, F. Sheng, X. Du, The mediator subunit OsMED16 interacts with the WRKY transcription factor OsWRKY45 to enhance rice resistance against *Magnaporthe oryzae*, Rice 17 (2024) 23.
[158] T. Wei, B. Ou, J. Li, Y. Zhao, D. Guo, Y. Zhu, Z. Chen, H. Gu, C. Li, G. Qin, Transcriptional profiling of rice early response to Magnaporthe oryzae identified OsWRKYs as important regulators in rice blast resistance, PLoS ONE 8 (2013) e59720.
[159] S.H. Hwang, S.I. Kwon, J.Y. Jang, I.L. Fang, H. Lee, C. Choi, S. Park, I. Ahn, S.C. Bae, D.J. Hwang, OsWRKY51, a rice transcription factor, functions as a positive regulator in defense response against *Xanthomonas oryzae* pv. *oryzae*, Plant Cell Rep. 35 (2016) 1975–1985.
[160] T. Chujo, R. Takai, C. Akimoto-Tomiyama, S. Ando, E. Minami, Y. Nagamura, H. Kaku, N. Shibuya, M. Yasuda, H. Nakashita, K. Umemura, A. Okada, K. Okada, H. Nojiri, H. Yamane, Involvement of the elicitor-induced gene OsWRKY53 in the expression of defense-related genes in rice, Biochim. Biophys. Acta 1769 (2007) 497–505.
[161] T. Chujo, K. Miyamoto, S. Ogawa, Y. Masuda, T. Shimizu, M. Kishi-Kaboshi, A. Takahashi, Y. Nishizawa, E. Minami, H. Nojiri, H. Yamane, K. Okada, Overexpression of phosphomimic mutated OsWRKY53 leads to enhanced blast resistance in rice, PLoS ONE 9 (2014) e98737.
[162] S.J. Yoo, S.H. Kim, M.J. Kim, C.M. Ryu, Y.C. Kim, B.H. Cho, K.Y. Yang, Involvement of the OsMKK4-OsMPK1 cascade and its downstream transcription factor OsWRKY53 in the wounding response in rice, Plant Pathol. J. 30 (2014) 168–177.
[163] L. Hu, M. Ye, R. Li, T. Zhang, G. Zhou, Q. Wang, J. Lu, Y. Lou, The rice transcription factor WRKY53 suppresses herbivore-induced defenses by acting as a negative feedback modulator of mitogen-activated protein kinase activity, Plant Physiol. 169 (2015) 2907–2921.
[164] L. Hu, M. Ye, R. Li, Y. Lou, OsWRKY53, a versatile switch in regulating herbivore-induced defense responses in rice, Plant Signal. Behav. 11 (2016) e1169357.
[165] W. Xie, Y. Ke, J. Cao, S. Wang, M. Yuan, Knock out of transcription factor WRKY53 thickens sclerenchyma cell walls, confers bacterial blight resistance, Plant Physiol. 187 (2021) 1746–1761.
[166] J. Zhang, Y. Peng, Z. Guo, Constitutive expression of pathogen-inducible OsWRKY31 enhances disease resistance and affects root growth and auxin response in transgenic rice plants, Cell Res. 18 (2008) 508–521.
[167] S. Wang, S. Han, X. Zhou, C. Zhao, L. Guo, J. Zhang, F. Liu, Q. Huo, W. Zhao, Z. Guo, X. Chen, Phosphorylation and ubiquitination of OsWRKY31 are integral to OsMKK10-2-mediated defense responses in rice, Plant Cell 35 (2023) 2391–2412.
[168] Y. Peng, L.E. Bartley, X. Chen, C. Dardick, M. Chern, R. Ruan, P.E. Canlas, P.C. Ronald, OsWRKY62 is a negative regulator of basal and Xa21-mediated defense against *Xanthomonas oryzae* pv. *oryzae* in rice, Mol. Plant 1 (2008) 446–458.
[169] C.J. Park, P.C. Ronald, Cleavage and nuclear localization of the rice XA21 immune receptor, Nat Commun. 3 (2012) 920.
[170] Z. Zhu, T. Wang, J. Lan, J. Ma, H. Xu, Z. Yang, Y. Guo, Y. Chen, J. Zhang, S. Dou, M. Yang, L. Li, G. Liu, Rice MPK17 plays a negative role in the Xa21-mediated resistance against *Xanthomonas oryzae* pv. *oryzae*, Rice 15 (2022) 41.
[171] X. Shi, Y. Xiong, K. Zhang, Y. Zhang, J. Zhang, L. Zhang, Y. Xiao, G.L. Wang, W. Liu, The ANIP1-OsWRKY62 module regulates both basal defense and Pi9-mediated immunity against *Magnaporthe oryzae* in rice, Mol. Plant 16 (2023) 739–755.
[172] Z.W. Fu, J.H. Li, X. Gao, S.J. Wang, T.T. Yuan, Y.T. Lu, Pathogen-induced methylglyoxal negatively regulates rice bacterial blight resistance by inhibiting OsCDR1 protease activity, Mol. Plant 17 (2024) 325–341.
[173] Y.S. Seo, M. Chern, L.E. Bartley, M. Han, K.H. Jung, I. Lee, H. Walia, T. Richter, X. Xu, P. Cao, W. Bai, R. Ramanan, F. Amonpant, L. Arul, P.E. Canlas, R. Ruan, C.J. Park, X. Chen, S. Hwang, J.S. Jeon, P.C. Ronald, Towards establishment of a rice stress response interactome, PLoS Genet. 7 (2011) e1002020.
[174] J. Liu, X. Chen, X. Liang, X. Zhou, F. Yang, J. Liu, S.Y. He, Z. Guo, Alternative splicing of rice WRKY62 and WRKY76 transcription factor genes in pathogen defense, Plant Physiol. 171 (2016) 1427–1442.
[175] X. Liang, X. Chen, C. Li, J. Fan, Z. Guo, Metabolic and transcriptional alternations for defense by interfering OsWRKY62 and OsWRKY76 transcriptions in rice, Sci Rep. 7 (2017) 2474.
[176] X. Xu, H. Wang, J. Liu, S. Han, M. Lin, Z. Guo, X. Chen, OsWRKY62 and OsWRKY76 interact with importin alpha1s for negative regulation of defensive responses in rice nucleus, Rice 15 (2022) 12.
[177] S. Son, G. Song, S. Nam, J. Lee, D.J. Hwang, E.J. Suh, S.R. Park, OsWRKY65 enhances immunity against fungal and bacterial pathogens in rice, Crop J. 12 (2024) 470–481.
[178] K.T.X. Vo, C.Y. Kim, T.V. Hoang, S.K. Lee, G. Shirsekar, Y.S. Seo, S.W. Lee, G.L. Wang, J.S. Jeon, OsWRKY67 plays a positive role in basal and XA21-mediated resistance in rice, Front. Plant Sci. 8 (2017) 2220.

[179] Q. Liu, X. Li, S. Yan, T. Yu, J. Yang, J. Dong, S. Zhang, J. Zhao, T. Yang, X. Mao, X. Zhu, B. Liu, OsWRKY67 positively regulates blast and bacteria blight resistance by direct activation of PR genes in rice, BMC Plant Biol. 18 (2018) 257.
[180] Y. Zhao, X. Zhu, C.M. Shi, G. Xu, S. Zuo, Y. Shi, W. Cao, H. Kang, W. Liu, R. Wang, Y. Ning, G.L. Wang, X. Wang, OsEIL2 balances rice immune responses against (hemi)biotrophic and necrotrophic pathogens via the salicylic acid and jasmonic acid synergism, New Phytol. (2024) 362–380.
[181] Z. Zhu, T.X.Z. Wang, Y. Chen, Y.Q. Liu, G.-W. Yan, S. Xu, J.J. Ma, S.J. Dou, L.Y. Li, G.Z. Liu, Rice transcription factor WRKY68 plays a positive role in Xa21-mediated resistance to *Xanthomonas oryzae* pv. *oryzae*, Acta Agron. Sin. 48 (2022) 1129–1140.
[182] X. Liu, X. Bai, X. Wang, C. Chu, OsWRKY71, a rice transcription factor, is involved in rice defense response, J. Plant Physiol. 164 (2007) 969–979.
[183] T. Chujo, T. Kato, K. Yamada, R. Takai, C. Akimoto-Tomiyama, E. Minami, Y. Nagamura, N. Shibuya, M. Yasuda, H. Nakashita, K. Umemura, A. Okada, K. Okada, H. Nojiri, H. Yamane, Characterization of an elicitor-induced rice WRKY gene, OsWRKY71, Biosci. Biotechnol. Biochem. 72 (2008) 240–245.
[184] X. Li, J. Zhang, X. Shangguan, J. Yin, L. Zhu, J. Hu, B. Du, W. Lv, Knockout of OsWRKY71 impairs Bph15-mediated resistance against brown planthopper in rice, Front. Plant Sci. 14 (2023) 1260526.
[185] Y. Hou, Y. Wang, L. Tang, X. Tong, L. Wang, L. Liu, S. Huang, J. Zhang, SAPK10-mediated phosphorylation on WRKY72 releases its suppression on jasmonic acid biosynthesis and bacterial blight resistance, iScience 16 (2019) 499–510.
[186] N. Yokotani, Y. Sato, S. Tanabe, T. Chujo, T. Shimizu, K. Okada, H. Yamane, M. Shimono, S. Sugano, H. Takatsuji, H. Kaku, E. Minami, Y. Nishizawa, WRKY76 is a rice transcriptional repressor playing opposite roles in blast disease resistance and cold stress tolerance, J. Exp. Bot. 64 (2013) 5085–5097.
[187] N.Y. Choi, E. Lee, S.G. Lee, C.H. Choi, S.R. Park, I. Ahn, S.C. Bae, C.H. Hwang, D.J. Hwang, Genome-wide expression profiling of OsWRKY superfamily genes during infection with *Xanthomonas oryzae* pv. *oryzae* using real-time PCR, Front. Plant Sci. 8 (2017) 1628.
[188] Y. Li, S. Liao, P. Mei, Y. Pan, Y. Zhang, X. Zheng, Y. Xie, Y. Miao, OsWRKY93 dually functions between leaf senescence and in response to biotic stress in rice, Front. Plant Sci. 12 (2021) 643011.
[189] H. Wang, J. Hao, X. Chen, Z. Hao, X. Wang, Y. Lou, Y. Peng, Z. Guo, Overexpression of rice WRKY89 enhances ultraviolet B tolerance and disease resistance in rice plants, Plant Mol. Biol. 65 (2007) 799–815.
[190] S. Son, H.K. An, Y.J. Seol, S.R. Park, J.H. Im, Rice transcription factor WRKY114 directly regulates the expression of OsPR1a and Chitinase to enhance resistance against *Xanthomonas oryzae* pv. *oryzae*, Biochem. Biophys. Res. Commun. 533 (2020) 1262–1268.
[191] S. Son, J.H. Im, G. Song, S. Nam, S.R. Park, OsWRKY114 inhibits ABA-induced susceptibility to *Xanthomonas oryzae* pv. *oryzae* in rice, Int. J. Mol. Sci. 23 (2022) 8825.
[192] G. Song, S. Son, S. Nam, E.-J. Suh, S.I. Lee, S.R. Park, OsWRKY114 is a player in rice immunity against *Fusarium fujikuroi*, Int. J. Mol. Sci. 24 (2023) 6604.
[193] S. Son, S.R. Park, Plant translational reprogramming for stress resilience, Front. Plant Sci. 14 (2023) 1151587.
[194] Q. Wang, J. Li, L. Hu, T. Zhang, G. Zhang, Y. Lou, OsMPK3 positively regulates the JA signaling pathway and plant resistance to a chewing herbivore in rice, Plant Cell Rep. 32 (2013) 1075–1084.
[195] G. Zhou, J. Qi, N. Ren, J. Cheng, M. Erb, B. Mao, Y. Lou, Silencing OsHI-LOX makes rice more susceptible to chewing herbivores, but enhances resistance to a phloem feeder, Plant J. 60 (2009) 638–648.
[196] J. Lu, J. Li, H. Ju, X. Liu, M. Erb, X. Wang, Y. Lou, Contrasting effects of ethylene biosynthesis on induced plant resistance against a chewing and a piercing-sucking herbivore in rice, Mol. Plant 7 (2014) 1670–1682.
[197] S.C. Koo, B.C. Moon, J.K. Kim, C.Y. Kim, S.J. Sung, M.C. Kim, M.J. Cho, Y.H. Cheong, OsBWMK1 mediates SA-dependent defense responses by activating the transcription factor OsWRKY33, Biochem. Biophys. Res. Commun. 387 (2009) 365–370.
[198] M.F. Ercoli, D.D. Luu, E.Y. Rim, A. Shigenaga, A. Teixeira de Araujo, Jr., M. Chern, R. Jain, R. Ruan, A. Joe, V. Stewart, P. Ronald, Plant immunity: Rice XA21-mediated resistance to bacterial infection, Proc. Natl. Acad. Sci. U. S. A. 119 (2022) e2121568119.
[199] C. Ji, Z. Ji, B. Liu, H. Cheng, H. Liu, S. Liu, B. Yang, G. Chen, Xa1 allelic R genes activate rice blight resistance suppressed by interfering TAL effectors, Plant Commun. 1 (2020) 100087.
[200] C. Lim, K. Kang, Y. Shim, S.C. Yoo, N.C. Paek, Inactivating transcription factor OsWRKY5 enhances drought tolerance through abscisic acid signaling pathways, Plant Physiol. 188 (2022) 1900–1916.
[201] C. Wasternack, M. Strnad, Jasmonates: news on occurrence, biosynthesis, metabolism and action of an ancient group of signaling compounds, Int. J. Mol. Sci. 19 (2018) 697.
[202] C. Lim, K. Kang, Y. Shim, Y. Sakuraba, G. An, N.C. Paek, Rice ETHYLENE RESPONSE FACTOR 101 promotes leaf senescence through jasmonic acid-mediated regulation of OsNAP and OsMYC2, Front. Plant Sci. 11 (2020) 1096.